Language Models for Analysing Social Media Discourse: Resources and Applications

Thesis submitted in accordance with the requirements of Cardiff University for the degree of Doctor in Philosophy by

Dimosthenis Antypas

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Declaration

I hereby declare that, this thesis (and the work presented in it) is submitted to the School of Computer Science & Informatics, Cardiff University, in fulfilment of the requirements for the degree of Doctor of Philosophy, and that this thesis is entirely my own work.

Cardiff, ADD DATE

Dimosthenis Antypas

Abstract

Social media has become an integral part of daily life for millions worldwide, serving as a platform where people exchange opinions, express concerns, and influence one another. From natural disasters to political debates, social media platforms like X (formerly Twitter) and Reddit have become primary sources of news and public discourse. This activity results in the generation of vast amounts of data each day which, due to their sheer volume and largely unstructured nature, researchers and analysts face significant challenges in effectively studying and analysing it.

To analyse this complex data source, researchers frequently utilise Natural Language Processing (NLP) tools and techniques. However, social media data poses unique challenges distinct from more formal or structured text. The landscape of social media is constantly evolving, with new slang, trending topics, emojis, and brevity of posts creating additional hurdles that traditional NLP methods often struggle to address.

The emergence of Large Language Models (LLMs), such as GPT-3 (Brown et al., 2020a), has transformed the landscape of natural language processing, making it possible to analyse and generate text at an unprecedented scale. However, our findings indicate that smaller, specialized models can often perform competitively with, and occasionally outperform, these larger models on domain-specific tasks. significant when considering the resource and environmental costs associated with training and deploying large-scale models, underscoring the importance of efficient, task-oriented model development.

In this work, we introduce several domain-specific models and datasets tailored specifically for social media data. We apply these specialized tools to studies focusing on politically charged content, hate speech detection, and misinformation. By leveraging these resources, we aim to demonstrate that specialized models not only offer competitive performance but also provide a more sustainable approach to social media data analysis.

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Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Pretraining Approach
ELMo	Embeddings from Language Models
BoW	Bag of Words
ICL	in-context learning
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
LM	Language Model
MLM	Masked Language Model

PLM	Pre-trained Language Model
ML	Machine Learning
NLP	Natural Language Processing
SoTA	State of The Art
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
URL	Uniform Resource Locator
MTLD	Measure of Textual Lexical Diversity
GloVe	Global Vectors for Word Representation
T5	Text-to-Text Transfer Transformer
BART	Bidirectional and Auto-Regressive Transformers
Flan-T5	Fine-tuned Language Net T5
C4	Colossal Clean Crawled Corpus
QLoRA	Quantized Low-Rank Adaptation
NER	Named Entity Recognition
LLaMA	Large Language Model Meta AI
NMF	Non-negative Matrix Factorization

1 Introduction

The rise of social media platforms have transformed the way people communicate and interact online. Platforms like *Twitter* and *Reddit*, have grown to be an integral part of the lives of millions of people, with approximately 63% of the population using a social media platform daily [1], and are seen as key sources of real-time information, public opinion, and social dynamics, making them valuable data sources for understanding human behavior and societal trends (Barberá, 2015; Shu et al., 2017). As populations shape opinions and influence each other through these platforms (Ausat, 2023), it has become crucial, from both business and scientific perspectives, to better understand how people interact and what factors drive their behaviors online.

This need becomes even more apparent in the context of political discourse, given how easily voters can be influenced and manipulated through targeted social media campaigns Goodwin et al. (2023). For example, Zhang et al. (2025) demonstrate how President Trump's often incendiary posts on Twitter and Truth Social amplified social media user engagement and attracted increased news media

[1] https://www.statista.com/statistics/617136/digital-population-worldwide/

attention. Similar studies on other major political events, such as Brexit (Mora-Cantallops et al., 2021), highlight the pivotal role social media platforms play in shaping how information spreads through networks, facilitating content generation, and influencing user behaviour.

To effectively leverage this rich but chaotic data, NLP provides an essential set of tools for extracting, structuring, and analyzing textual content from social media. Standard tasks include sentiment analysis(Zimbra et al., 2018), topic modeling (Egger and Yu, 2022), and entity recognition (Nasar et al., 2021), which help quantify emotional tone, uncover underlying themes, and identify key actors within discussions. Recent advances in neural architectures, such as transformers (Subakan et al., 2021), enable deeper contextual understanding and have paved the way for more nuanced tasks, including stance detection (AlDayel and Magdy, 2021), misinformation tracking (Hayawi et al., 2022), and aspect-based sentiment analysis (Brauwers and Frasincar, 2022). These methods allow researchers to convert raw, unstructured posts into structured formats that support downstream analysis.

Furthermore, social media data allows for tracking of user behaviour, discourse evolution, and community dynamics (Feuerriegel et al., 2025; Arazzi et al., 2023). Through the use of NLP techniques it becomes possible to detect coordinated campaigns, assess the spread of narratives, or evaluate public reactions over time (Luceri et al., 2024). These insights are increasingly valuable in fields such as computational social science, marketing, and policy analysis, where understanding the connections between language, behaviour, and influence is crucial. The intersection of NLP and social media offers significant opportunities for both methodological innovation and impactful, real-world applications.

Despite their potential, analysing social media data poses numerous challenges for

the field of NLP. User-generated content on these platforms often lacks context and is highly informal, containing slang, emojis, abbreviations, typos, and other forms of noisy data (Eisenstein, 2013; Gain et al., 2023). Moreover, language usage on social media evolves rapidly, making it difficult for static models and lexicons to keep pace. These characteristics demand robust, adaptive, and context-aware approaches to effectively process and interpret the vast and dynamic stream of social media text.

In order to tackle the limitations above, we explore existing methodologies and resources that work on understanding complex social interactions on social media. Subsequently, we introduce new models and datasets that improve on previous work, apply them on large corpora collected from *Twitter* and *Reddit* to understand better human interactions regarding politics discourse, and misinformation and hate speech proliferation. The rest of this introductory chapter is organised as follows: § 1.1 outlines the research aim and motivations for this thesis, § 1.2 outlines the contribution of research, § 1.3 lists publications that are peer-reviewed or currently under review, and § 1.6 describes the structure of the thesis and summarises this introductory chapter.

1.1 Motivation

NLP applications are essential for understanding human interactions in social media, especially in the context of political discourse and dissemination of misinformation and hate speech. For example, Machine Learning (ML) models are used by social platforms, such as Facebook (Schroepfer, 2021), to detect harmful content and protect users from online abuse. Similarly, researchers have developed tools to analyse political communication on Twitter (Barberá, 2015), track the spread of misinformation (Shu et al., 2017), and understand the dynamics of online discussions (Zhang et al., 2018).

Many hate speech detection models rely on supervised learning techniques, which require large amounts of annotated data to train effectively. However, the availability of high-quality annotated datasets for hate speech detection is limited, and existing datasets often suffer from biases and inconsistencies (Davidson et al., 2017). This can lead to models that are not generalisable across different contexts or languages, limiting their effectiveness in real-world applications. Other applications, such as misinformation detection and sentiment analysis, also face similar challenges in terms of data availability and quality. For example, misinformation detection models often rely on large-scale datasets of social media posts, which can be difficult to obtain and may not accurately reflect the diversity of online content (Shu et al., 2017). Similarly, sentiment analysis models often struggle to capture the nuances of social media language, which can vary significantly across different platforms and user demographics (Barbieri et al., 2022).

Due to the heavily unstructured content often shared in social media content, many of the aforementioned applications rely on specialised datasets to perform well. These datasets are often collected from social media platforms, such as *Twitter* and *Reddit*, and annotated by human coders to provide ground truth labels for training and evaluation. However, the quality and reliability of these datasets can vary significantly, depending on the annotation process and the expertise of the annotators (Klie et al., 2024). This can lead to biases and inaccuracies in the models trained on these datasets, limiting their generalisability and effectiveness in real-world scenarios.

The proliferation of social media platforms has created unprecedented challenges in political discourse and misinformation spread (Muhammed T and Mathew, 2022). Political polarization and the rapid dissemination of false information have become major societal concerns, affecting democratic processes and public opinion formation (Allcott and Gentzkow, 2017). These challenges are further complicated by the heavily unstructured nature of social media content, which can pose difficulties even for larger models (Eisenstein, 2013; Gain et al., 2023). Finally, it is crucial to recognise both the potential biases in existing models, tools, and datasets, as well as the environmental impact associated with developing and deploying LLMs (Bender et al., 2021).

1.2 Contributions

The main motivation of this thesis is to explore how NLP techniques can be effectively applied to social media data in order to address complex societal questions. Social media platforms generate vast amounts of unstructured textual

data that offer unique insights into public opinion, communication patterns, and societal events. However, leveraging this data poses significant challenges due to its informal language, high variability, and noise. This raises a central research question: How can NLP methodologies be adapted or developed to robustly analyse and interpret social media content for meaningful societal insights? This question branches into more specific problems concerning data representation, model design, annotation strategies, and evaluation practices.

Rather than attempting to solve all aspects of the problem, this thesis aims to fill gaps in existing literature on concrete subproblems where progress can be made on how to detect subtle patterns in public discourse, how to model social dynamics from textual evidence, and how to evaluate such models in realistic settings. Specifically, the research presented in this thesis focuses on two core areas: (1) developing and evaluating NLP resources for social media analysis; and (2) applying these resources through well-defined approaches to analyse social media dynamics in the context of politics discourse, misinformation spread and hate speech propagation. By addressing these areas, the thesis aims to provide new insights into the dynamics of social media communication and contribute to the development of more effective tools for understanding and combating harmful online behaviours. To this end, the thesis makes a number of noticeable contributions as listed below:

1.2.1 Part 1: Modeling Approaches for Social Media Language Understanding

The first core contribution of the thesis focuses on developing and evaluating NLP tools specifically designed for social media applications.

I.1 **New Datasets for Topic Classification on Twitter.** Extracting meaningful insights from the vast amount of data generated on Twitter is inherently challenging, and performing targeted topic analysis adds further complexity. Unsupervised methods such as topic modelling are often difficult to evaluate and interpret, while existing datasets for supervised classification are not well-suited for the informal and dynamic nature of social media content. To address this gap, we introduce two novel datasets, TweetTopic and it is multilingual extension X-Topic, specifically designed for training and evaluating topic classification models on Twitter data. These datasets, along with corresponding State of The Art (SoTA) models, are detailed in Chapters 3 and **??**.

I.2 **Unification of Hate Speech Detection Datasets.** We present a unified hate speech detection dataset specifically curated for social media content by leveraging and combining multiple existing resources. Although hate speech detection is a widely studied problem, many available datasets lack diversity and tend to focus on specific target groups, limiting the generalisability of trained models. By systematically aggregating and harmonising these resources, we construct a comprehensive dataset that, as demonstrated in **??**, enables models to achieve superior performance compared to those trained on individual datasets.

I.3 **Introduction of a Benchmark for Twitter-related Tasks.** We present SuperTweetEval, a benchmark that unifies and standardises the evaluation of social media NLP tasks, addressing the current fragmentation in this area. Comprising XX diverse tasks, our benchmark poses significant challenges not only for smaller Pre-trained Language Models (PLMs) but also for the latest LLMs, as discussed in **??**.

A common observation across all three contributions is that smaller, specialised models, such as RoBERTa, can perform as well as, and often outperform, larger LLMs like GPT-4 on the evaluated tasks when fine-tuned. This observation is primarily due to the specialised training data: the models we evaluated are frequently trained on large corpora of social media content, making them better suited for domain-specific language use and nuances. Additionally, their architectural design contributes to this advantage. Unlike generative LLMs, which are based on a decoder architecture, smaller PLMs typically employ an encoder architecture, which is inherently more effective for discriminative tasks such as classification. Encoders focus on learning contextual representations of input sequences, which allows for precise feature extraction and robust classification performance which is particularly important for tasks like topic classification and hate speech detection, where subtle linguistic cues and contextual dependencies are critical.

The impact of this thesis is evidenced by the widespread adoption of the released datasets and models. For instance, the topic classification resources introduced in Chapters 3 and **??** have accumulated over 1,200,000 downloads on the Hugging Face platform[2], while our hate speech classifier (**??**) has more than 1,400,000

[2] https://huggingface.co/collections/cardiffnlp/tweettopic-65eb2a0eada92

downloads[3], highlighting their practical value and community uptake.

Apart from the wide adoption of the resources, the models and datasets have been used in various studies from different domains. For example, they have informed research in tourism studies, where Lee et al. (2025) employed our topic classification model to perform a targeted investigation on online hostility directed at tourists in the context of over-tourism debates. Similarly, our hate speech classifier (**??**) has demonstrated performance comparable to, or surpassing, established tools such as Google's Perspective API[4] and OpenAI's Moderation system[5], as shown by Shen et al. (2025). It has also been adopted by researchers to examine the temporal dynamics of hate speech on Twitter Arun et al. (2024), and to investigate correlations between hate speech and crime statistics (Bicakci et al., 2025).

This interdisciplinary application underscores the utility of robust NLP tools in social science and humanities contexts, as evidenced by citations our papers have achieved across computer science, linguistics, political science, psychology, business, environmental science, medicine, and philosophy[6].

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[3] https://huggingface.co/cardiffnlp/twitter-roberta-base-hate-latest/settings

[4] https://perspectiveapi.com/

[5] https://platform.openai.com/docs/guides/moderation

[6] According to data from https://www.semanticscholar.org/

1.2.2 Part 2: Analysing Harmful and Political Dynamics in Social Media

We use the resources developed and insights extracted in the previous chapters to analyse complex social media dynamics.

II.1 **Politics and Negativity.** We conduct a large-scale analysis of posts by politicians from eight different parliaments across five languages (including English, Spanish, and Greek) to investigate how negatively charged tweets tend to spread more widely and rapidly than neutral or positive ones. To achieve this, we train domain-specific sentiment analysis models and develop a systematic approach to quantify and track the diffusion of such content. Further details are provided in **??**.

II.2 **Profiling Misinformation Spreaders.** We present a methodology to profile users who frequently share misinformation on social media. By leveraging our previously developed models for topic classification and hate speech detection, along with sentiment and emotion analysis, we systematically analyse a large number of user accounts to uncover behavioural patterns and content characteristics associated with misinformation spread. Further details are provided in **??**.

II.3 **Trigger Points Analysis.** Trigger points are a relatively new concept referring to specific words or phrases that can provoke strong emotional reactions in discourse. We propose an approach for systematically identifying potential trigger points using various lexical features, focusing mainly on sentiment, emotion, and the presence of hate speech. Unlike previous

qualitative studies that introduced the notion of trigger points, our approach adopts a quantitative perspective. Further details are provided in **??**.

Throughout this work, we provide detailed and transparent methodologies and approaches that are not only robust but also easy to replicate and adapt. The approaches we introduce are designed to be easily applicable to other datasets and platforms beyond Twitter and Reddit, enabling researchers and practitioners to extend our analyses to a wide range of social media sources and related contexts.

Finally, it is worth noting that there was a consistent effort to collaborate with experts from other fields, such as social sciences and healthcare, and to incorporate their feedback in order to address real-world problems 'through our research. These interdisciplinary collaborations, as seen in § 1.3.2.1, highlight the practical relevance of our work.

1.3 Publications

The majority of the material in this thesis have been published in peer-reviewed conferences and journals, while the remaining material are under reviewed at peer-reviewed conferences. Below is a list of the core publications where the main contributions of the thesis have been published in the chronological order.

1.3.1 Core Publications

Below is a list of the core publications where the main contributions of the thesis have been published in chronological order. The majority of these the publications have been led by myself. However, in two cases, pecifically publications (1) and (7), the second co-author made substantial contributions. In publication (1), Asahi Ushio was responsible for collecting the raw data and designing the annotation guidelines used to create the TweetTopic dataset (Chapter 3). In publication (7), Christian Arnold developed the statistical framework employed to examine the effect of trigger points (**??**).

1.3.1.1 Part 1: Modeling Approaches for Social Media Language Understanding

1. Dimosthenis Antypas, Asahi Ushio, Jose Camacho-Collados, Vitor Silva, Leonardo Neves, and Francesco Barbieri. 2022. Twitter Topic Classification. In Proceedings of the 29th International Conference on Computational Linguistics, pages 3386–3400, Gyeongju, Republic of Korea. International Committee on Computational Linguistics. Chapter 3

2. Dimosthenis Antypas, Asahi Ushio, Francesco Barbieri, and Jose Camacho-Collados. 2024. Multilingual Topic Classification in X: Dataset and Analysis. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 20136–20152, Miami, Florida, USA. Association for Computational Linguistics. **??**

3. Dimosthenis Antypas and Jose Camacho-Collados. 2023. Robust Hate Speech

Detection in Social Media: A Cross-Dataset Empirical Evaluation. In The 7th Workshop on Online Abuse and Harms (WOAH), pages 231–242, Toronto, Canada. Association for Computational Linguistics. **??**

4. Dimosthenis Antypas, Asahi Ushio, Francesco Barbieri, Leonardo Neves, Kiamehr Rezaee, Luis Espinosa-Anke, Jiaxin Pei, and Jose Camacho-Collados. 2023. SuperTweetEval: A Challenging, Unified and Heterogeneous Benchmark for Social Media NLP Research. In Findings of the Association for Computational Linguistics: EMNLP 2023, pages 12590–12607, Singapore. Association for Computational Linguistics. **??**

1.3.1.2 Part 2: Analysing Harmful and Political Dynamics in Social Media

5. Dimosthenis Antypas, Alun Preece, and Jose Camacho-Collados. 2023. Negativity spreads faster: A large-scale multilingual Twitter analysis on the role of sentiment in political communication. Online Social Networks and Media, 33:100242. **??**

6. Dimosthenis Antypas, Alun Preece, and Jose Camacho-Collados. 2024. A Multi-Faceted NLP Analysis of Misinformation Spreaders in Twitter. In Proceedings of the 14th Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis, pages 71–83, Bangkok, Thailand. Association for Computational Linguistics. **??**

7. Dimosthenis Antypas, Christian Arnold, Jose Camacho-Collados, Nedjma Ousidhoum, and Carla Perez Almendros. 2024. Words as Trigger Points in Social Media Discussions. *Under Review in ICSWM 2025.* **??**

1.3.2 Other Publications

The following publications are not formally part of this thesis. However, they illustrate the applicability of the research in diverse real-world scenarios, particularly in handling the nuances of social media data. They also highlight the interdisciplinary relevance of our work as they connect NLP with social sciences, health care, and disaster management.

1.3.2.1 Collaboration with public stakeholders

During the duration of this thesis there were opportunities to collaborate with various public stakeholders addressing challenges arising from social media usage. These collaborations resulted in several publications, showcasing the real-world applicability of our methods.

The first paper (8) emerged from a collaboration with the Spanish National Office Against Hate Crimes. It's aim was to examine potential temporal trends in hate speech on Twitter, particularly around Spanish national holidays (e.g., Labour Day). My role focused on adapting our existing hate speech detection models to achieve robust performance in the Spanish language.

The second paper (9) was produced in collaboration with Public Health Wales. This work aimed to develop an approach for early detection of influenza outbreaks using Twitter data. My primary contribution involved curating suitable datasets and training RoBERTa-based models capable of identifying tweets that reported influenza-like symptoms, either from a first-person perspective or in general.

Another collaboration (10) included colleagues from the School of Earth and Environmental Sciences and aimed to provide post-disaster recovery assessment in the case of earthquakes by utilising sentiment analysis on relevant tweets. Using our expertise on the area we managed to train tailored and domain specific classifiers which out performed the initial, paid, service that was being used for sentiment analysis.

Finally, a collaboration with researchers from Snap Inc. (11) resulted in the development and public release of a dataset specifically designed for the detection of sensitive content on social media, along with accompanying detection models. My primary contributions to this project involved coordinating the collection and aggregation of annotations, as well as training and evaluating the Language Models (LMs) employed in the study.

8. Jesús Gómez, Alberto Matilla-Molina, Ma Pilar Amado, Dimosthenis Antypas, Jose Camacho-Collados, Carlos J. Máñez, Tomás Fernández-Villazala, Alicia Méndez-Sanchís, and Javier López. 2023. The interaction between offensive and hate speech on Twitter and relevant social events in Spain. In News Media and Hate Speech Promotion in Mediterranean Countries, pages 81–109. IGI Global.

9. Mark Drakesmith, Dimosthenis Antypas, Claire Brown, Jose Camacho-Collados, and Jiao Song. 2025. Towards a social media-based disease surveillance system for early detection of influenza-like illnesses: A Twitter case study in Wales. In Proceedings of the Tenth Workshop on Noisy and User-generated Text (W-NUT 2025).

10. Diana Contreras, Dimosthenis Antypas, Javier Hervas, Sean Wilkinson, Jose

Camacho-Collados, Philippe Garnier, and Cécile Cornou. 2025. Post-disaster recovery assessment using sentiment analysis of English-language tweets: A tenth-anniversary case study of the 2010 Haiti earthquake. Sustainability 17, no. 11: 4967.

11. Dimosthenis Antypas, Indira Sen, Carla Perez-Almendros, Jose Camacho-Collados, and Francesco Barbieri. 2025. Sensitive Content Classification in Social Media: A Holistic Resource and Evaluation. In The 9th Workshop on Online Abuse and Harms (WOAH), Vienna, Austria. Association for Computational Linguistics.

1.3.2.2 Complementary Studies on Social Media

In addition to collaborations with public stakeholders, several projects were undertaken that, while adjacent to the main goals of this thesis, are not part of its core contributions. Specifically, in work (12), we conducted a large-scale analysis of Twitter data, collecting a substantial corpus of tweets related to the COVID-19 pandemic. From this, we extracted and annotated a labelled dataset aimed at identifying pandemic-related misinformation and developed binary classifiers for misinformation detection.

Work (13) focused on classifying Twitter posts by geographic origin (European vs. non-European) and deriving insights from these groups using comparative techniques such as lexical specificity and word embedding models. My main contribution to this study involved the analysis of the collected corpus, particularly through the use of lexical specificity metrics (Camacho-Collados et al., 2016).

Work (14) utilised Reddit data, and aimed to develop classifiers capable of detecting early signs of depression in user posts. My involvement centered on exploring model architectures capable of handling long-form textual input, which was a significant challenge at the time.

Finally, work (15) was part of a large collaborative effort that resulted in the development of the TweetNLP Python library. This toolkit facilitates working with Twitter data by offering easy-to-use implementations for tasks such as sentiment analysis, topic classification, and question answering. My contributions primarily involved integrating models I had previously developed, such as those for topic classification and hate speech detection, into the library.

12. Dimosthenis Antypas, Jose Camacho-Collados, Alun Preece, and David Rogers. 2021. COVID-19 and Misinformation: A Large-Scale Lexical Analysis on Twitter. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing: Student Research Workshop, pages 119–126, Online. Association for Computational Linguistics.

13. David Tuxworth, Dimosthenis Antypas, Luis Espinosa-Anke, Jose Camacho-Collados, Alun Preece, and David Rogers. 2021. Deriving disinformation insights from geolocalized Twitter callouts. In Proceedings of KDD 2021 - Workshop on Deriving Insights From User-Generated Text.

14. David Owen, Dimosthenis Antypas, Athanasios Hassoulas, Antonio F. Pardiñas, Luis Espinosa-Anke, and Jose Camacho Collados. 2023. Enabling early health care intervention by detecting depression in users of web-based forums using language models: Longitudinal analysis and evaluation. JMIR

AI 2, no. 1.

15. Jose Camacho-collados, Kiamehr Rezaee, Talayeh Riahi, Asahi Ushio, Daniel Loureiro, Dimosthenis Antypas, Joanne Boisson, Luis Espinosa Anke, Fangyu Liu, and Eugenio Martínez Cámara. 2022. TweetNLP: Cutting-Edge Natural Language Processing for Social Media. In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pages 38–49, Abu Dhabi, UAE. Association for Computational Linguistics.

1.3.2.3 Other work

The previous work discussed has been largely adjacent to the core focus of this thesis, particularly with an emphasis on social media. In addition to this, I contributed to other projects that are less directly related. Specifically, in project (16), I was involved in the development of BLEnD, a benchmark designed to evaluate the cultural diversity of LLMs. This was achieved by introducing manually annotated question pairs covering everyday cultural knowledge from 16 countries and in 13 languages. My role focused on collecting and ensuring the quality of the Greek questions. In project (17), a large-scale study investigating how LMs process metaphors and analogies, I contributed to the creation of parts of the datasets used for model evaluation.

16. Myung, Junho, Nayeon Lee, Yi Zhou, Jiho Jin, Rifki Afina Putri, Dimosthenis Antypas, Hsuvas Borkakoty et al. "BLEnD: A Benchmark for LLMs on Everyday Knowledge in Diverse Cultures and Languages." *Under Review in NeurIPS 2024 Datasets and Benchmarks Track.*

17. Joanne Boisson, Asahi Ushio, Hsuvas Borkakoty, Kiamehr Rezaee, Dimosthenis Antypas, Zara Siddique, Nina White, and Jose Camacho-Collados. 2024. How Are Metaphors Processed by Language Models? The Case of Analogies. In Proceedings of the 28th Conference on Computational Natural Language Learning, pages 365–387, Miami, FL, USA. Association for Computational Linguistics.

1.4 Ethics and Privacy

The research included in the thesis has been approved by Cardiff University's ethics comitee and also follows the guidelines of ACL code of ethics[7].

Specifically, as our research focuses on socia media we have taken caution while collecting data used in our experiments to adhere to the aforementioned ethic rules. Firstly, all the data were collected using the official Application Programming Interfaces (APIs) provided by *Twitter* and *Reddit*. Moreover, the privacy of the users whose data is used is protected by masking user handles and Uniform Resource Locators (URLs) to prevent any potential identification of users. Exceptions are data colleted from verified[8] *Twitter* users which are already public figures such as politicians, celebrites, and goverment employees.

Finally, regarding the annotations conducted, all coders were paid at least minimum wage for their work, and no personal identifying information was collected. This applies to the manual data labeling for TweetTopicand X-Topic,

[7]https://www.acm.org/code-of-ethics

[8]As defined by the legacy verified policy of Twitter in https://help.x.com/en/managing-your-account/legacy-verification-policy.

as described in Chapters 3 and **??** respectively.

1.5 Naming conventions

Models. In Chapters **??** and **??**, the models used are described as SoTA. While this statement may not hold true at the time of writing this thesis, they were the best-performing models when the experiments were conducted.

Twitter. A lot of the data used were collected from *X* while it was known as *Twitter* through the use of it's now defunct academic API. In the thesis we will refer to the platform as *Twitter* to reflect the time of the data collection and research done. Exception being Chapter **??** where the data was collected from *X* and we will refer to the platform as *X*.

1.6 Thesis Outline and Summary

In this section, we provide an overview of the structure of the thesis and summarise the core themes of each chapter. Figure 1.1 shows the entire structure of this thesis, where chapters with same interest are grouped together and chapters that utilise tools are colored coded the same way.

The first set of chapters are preliminary chapters to review the literature of relational representation learning and LMs (Chapters 1 and 2). The second set of chapters focus on identifying limitations in existing resources related to social media NLP and proposing solutions. Mainly we present our study for topic

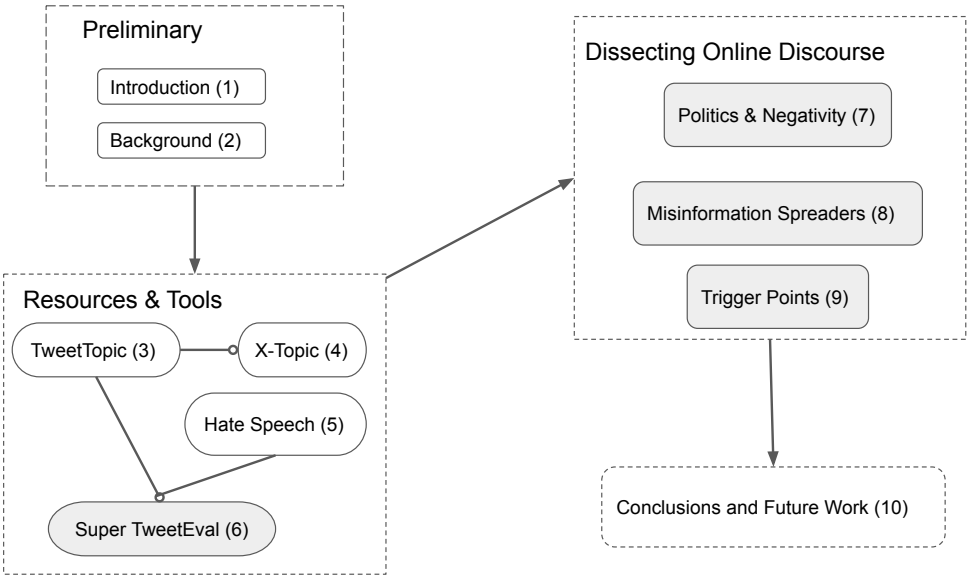

Figure 1.1: Outline of the thesis. The chapter number is indicated in brackets. The grey color indicates connections between chapters of Part 1 and Part 2.

classification in *Twitter* (Chapter 3), and our multilingual extension to this work (**??**). Following this we introduce an ensemble of models and datasets related to hate speech detection (**??**), and a benchmark for social media NLP tasks (**??**) which builds upon our previous work.

The final part of the thesis focuses on applying the aforementioned resources to understand complex social interactions in social media. **??** introduces a study on the role of sentiment in political communication, followed by an analysis of misinformation spreaders (**??**), and finally we investigate how specific words can act as trigger points in online discussions (**??**). Finally, we conclude the thesis with a summary of the contributions and future work (**??**).

As a temporal reference for this thesis, Figure 1.2 presents a high-level timeline of

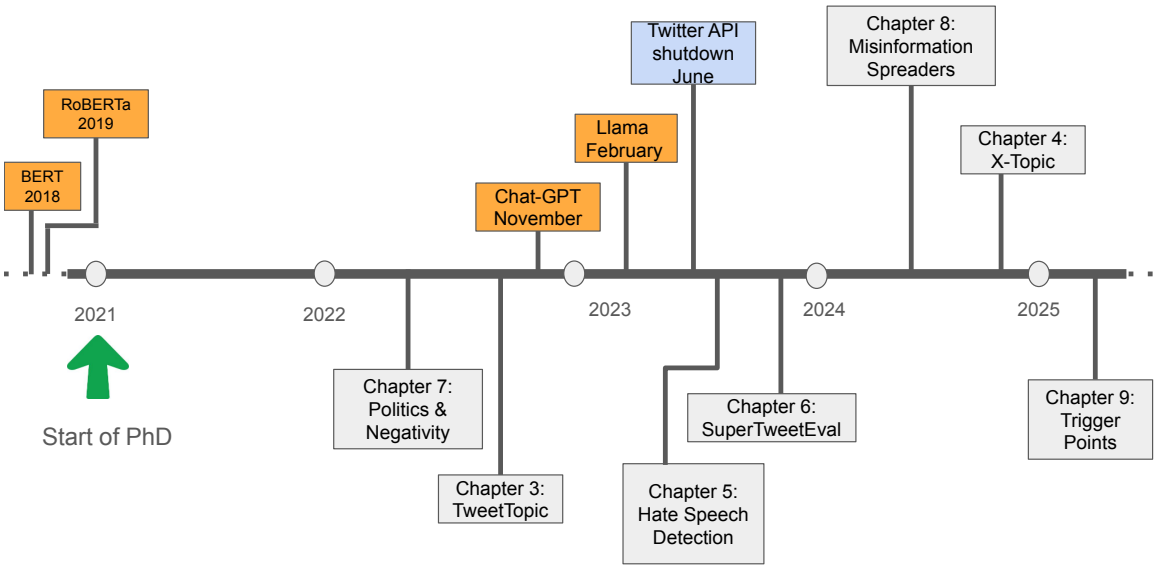

Figure 1.2: Timeline of papers publications along with major events in NLP field.

the publications that form part of this work, along with major events in the field of NLP, such as the release of key models and significant developments like the shutdown of the Twitter API. Although this thesis primarily focuses on utilising smaller LMs, the work conducted after the release of ChatGPT and open-source models like LLaMA incorporates results from these models and often compares their performance as well.

In the next chapter (Chapter 2), we will discuss the relevant literature and methodologies related to the research presented in this thesis. We will review existing work on key topics such as sentiment analysis, hate speech detection, misinformation analysis, and topic classification, and provide a brief, high-level introduction to computational models like LMs.

2 Background

This chapter provides the theoretical and methodological foundation of the thesis, covering the evolution of NLP tools and their applications to social media data. It explores the challenges of processing noisy, heterogeneous text from platforms like Twitter and discusses existing work on key areas such as hate speech detection, misinformation analysis, and multilingual topic classification. By situating the thesis within the broader research landscape, this chapter demonstrates how the presented contributions advance the state of the art.

2.1 Language Models

LMs are fundamental to NLP, aiming to capture the statistical properties of language and predict the probability of a sequence of words. Their evolution has significantly impacted the capabilities of NLP systems, especially in handling complex human language.

2.1.1 Traditional Language Models

2.1.1.1 Early Language Models

Early LMs were often based on statistical methods, such as N-gram models, which predict the next word based on the previous N-1 words (Buck et al., 2014). These models, while foundational, suffered from limitations in capturing long-range dependencies and semantic meaning. To represent words, early systems frequently relied on static word embeddings such as Word2Vec (Mikolov et al., 2013) and Global Vectors for Word Representation (GloVe) (Pennington et al., 2014), where each word was mapped to a single, fixed vector regardless of context. While these representations improved over one-hot encodings by capturing some semantic similarity (e.g., "king" and "queen" being close in vector space), they failed to reflect how a word's meaning could shift depending on its usage in different contexts.

These limitations were overcomed, up to a degree, through the development of neural approaches capable of modeling sequential dependencies and context more effectively. Specifically, Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTMs) and Gated Recurrent Units (GRUs), introduced a new class of language models capable of processing word sequences while maintaining a dynamic internal state (Yu et al., 2019; Dey and Salem, 2017). These models improved the contextual understanding of language over static embeddings by updating representations as each token in a sequence was processed. However, they still faced challenges in handling long-term dependencies and lacked efficient parallelization during training, which restricted scalability and performance

on large datasets (Sherstinsky, 2020).

2.1.1.2 Masked Language Models

Masked Language Models (MLMs) emerged with the introduction of the Transformer architecture (Subakan et al., 2021). Unlike traditional LMs that predict the next word, MLMs are trained by masking a percentage of the tokens in the input and then predicting those masked tokens based on the context provided by the unmasked tokens as seen in Figure 2.1 (Devlin et al., 2019). This bidirectional training allows them to learn richer, contextualized representations of words. These models are typically pretrained on massive corpora such as Wikipedia and the Common Crawl[1], enabling them to learn robust language representations.

For example, the embedding for the word bank in "he sat by the river bank" will differ significantly from that in "she went to the bank to deposit money," because the surrounding context informs the model about the intended sense of the word. This is in stark contrast to earlier approaches like Word2Vec or GloVe, which assign the same vector to bank regardless of its meaning in different contexts. These contextual embeddings allow MLMs to capture intricate semantic and syntactic relationships, making them highly effective across a wide range of downstream tasks.

Most contemporary MLMs are based on the BERT architecture, which is an encoder-based model leveraging bidirectional self-attention mechanisms to capture contextual information from both the left and right of each token (Devlin et al., 2019). Extensions to BERT include Robustly Optimized BERT Pretraining

[1] https://commoncrawl.org

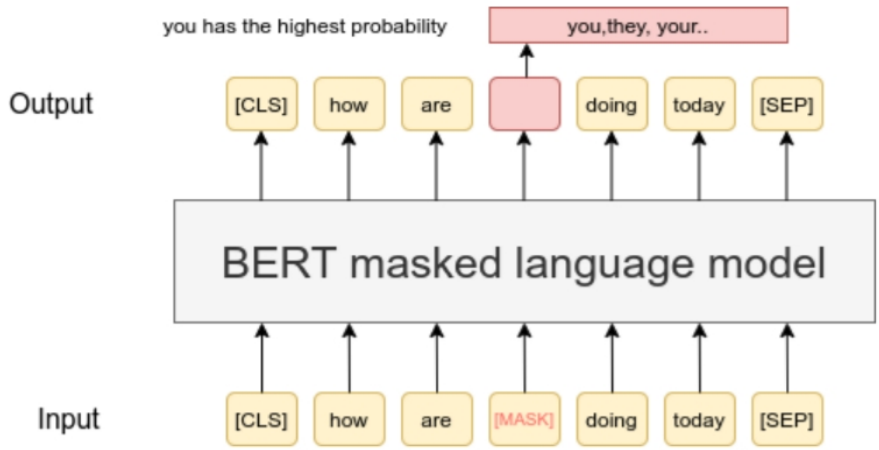

Figure 2.1: Example of MLM objective for training in Bidirectional Encoder Representations from Transformers (BERT).

Approach (RoBERTa), which optimises the pre-training process and introduces larger datasets, longer training, and XLM-R, which extends RoBERTa to multilingual settings (Liu et al., 2019a; Conneau et al., 2020).

One of the main benefits of these models is that their knowledge can be utilised in various downstream tasks with minimal modifications. Also, due to their encoder architecture which allows them to understand context from both left and right, they display significantly improved performance when fine-tuned on tasks like text classification, question answering, and named entity recognition (Yang et al., 2023).MLMs have become the backbone of many state-of-the-art NLP systems, enabling significant advancements in understanding and generating human language. They are particularly effective for tasks that require deep contextual understanding, such as sentiment analysis (Alaparthi and Mishra, 2021), hate speech detection (Mozafari et al., 2019), and misinformation detection (Kaliyar et al., 2021).

2.1.2 Encoder-Decoder Models

Encoder-decoder models represent a distinct approach in language modeling, particularly effective for sequence-to-sequence tasks. These architectures consist of two components: (i) an encoder that processes the input sequence and compresses its information into a latent representation, and (ii) a decoder that generates an output sequence based on this representation. This architecture is well-suited for tasks where the output differs in form or length from the input, such as translation, summarization, and question generation. Similar to MLMs, encoder-decoder models are also be pretrained on large corpora, while leveraging self-attention mechanisms in both the encoder and decoder stacks which allows them to learn bidirectional representations in the encoder while simultaneously training the decoder to generate sequences from noisy or incomplete inputs.

Encoder-decoder models became popular with the introduction of Text-to-Text Transfer Transformer (T5) (Raffel et al., 2020a). T5 casts NLP problems, such as summarisation, machine translation, and question asnwering, into a unified text-to-text format. It is pretrained on the Colossal Clean Crawled Corpus (C4) corpus(Raffel et al., 2020b) using a span corruption objective, where spans of text are masked and replaced with unique sentinel tokens, and the model learns to reconstruct the missing spans which enables it to develop robust representations for both understanding and generation (Figure 2.2).

An extension of T5, Fine-tuned Language Net T5 (Flan-T5) (Chung et al., 2024), further improves performance through instruction tuning (Wei et al., 2021) on a wide variety of tasks and datasets which enhances it's zero-shot and few-shot generalization capabilities. Another notable encoder-decoder model is Bidirectional

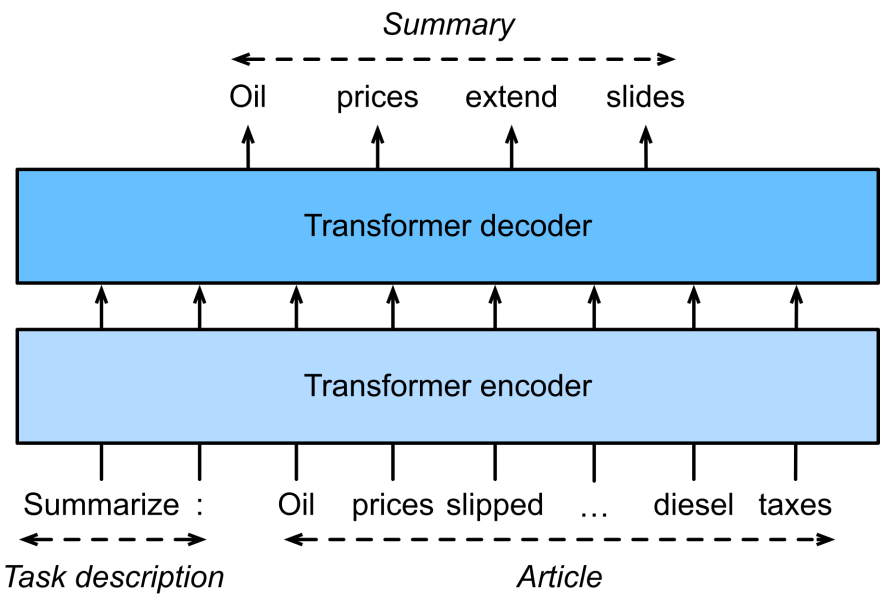

Figure 2.2: Example of T5 finetuning for a summarisation task.

and Auto-Regressive Transformers (BART) (Lewis et al., 2019), which is trained on a denoising autoencoder objectivewhere the model learns to reconstruct original text from corrupted input. BART combines the strengths of both MLMs and autoregressive models, making it effective for tasks like summarisation and translation.

Encoder-decoder models are widely used in tasks where the input and output differ significantly in structure or length, and similarly to MLMs they can be further finetuned for specific applications. Common use cases include machine translation, text summarization, question answering, and question generation. They can also be adapted for text generation tasks, producing coherent and contextually relevant text from prompts. However, these models are generally more complex and computationally intensive than MLMs, and they not always outperform encoder-only models on tasks that do not involve sequence generation (Gasparetto

et al., 2022). At the same time, they often lag behind decoder-only models, such as LLMs, in open-ended text generation tasks (Saattrup Nielsen et al., 2025).

2.1.3 Large Language Models

LLMs represent the most recent advancement in language modeling, building upon the foundations laid by MLMs and encoder-decoder models. They are characterized by their massive scale, often containing hundreds of billions of parameters, and their ability to perform a wide range of tasks with remarkable performamce, often in a zero-shot or few-shot learning setting (Wang et al., 2020). Unlike traditional LMs, which are typically trained on specific tasks or datasets, LLMs are pretrained on vast and diverse datasets encompassing text, code, and other modalities like images.

Architecturally, most LLMs are based on the Transformer decoder-only design and are trained using an autoregressive objective (Brown et al., 2020a)where, given a sequence of tokens, the model learns to predict the next token in a left-to-right manner. In contrast to bidirectional models like BERT, decoder-only models do not have access to future tokens during training, which makes them suited for generative tasks such as text completion, dialogue, summarization, and code generation.

Modern LLMs are capable to perform tasks based purely on natural language instructions, making *prompt engineering* a critical aspect of their effective use (Reynolds and McDonell, 2021). Moreover, similarly to smaller LMs their performance on specific tasks can often be further improved through finetuning. This is typically applied to smaller versions of the models using techniques like

Quantized Low-Rank Adaptation (QLoRA) (Dettmers et al., 2023), which enables efficient adaptation with limited computational resources.

Several notable LLMs have significantly advanced the field of NLP. For example, OpenAI's GPT series, starting with GPT-2 (Radford et al., 2019), demonstrated the potential of large-scale unsupervised pretraining and autoregressive text generation. Its succesors, GPT-3 (Brown et al., 2020b) and GPT-4 (Achiam et al., 2023) further scaled up the model size and training data, achieving impressive performance on a wide range of tasks with few-shot and zero-shot learning capabilities.

Alongside proprietary models, several open-sourced LLMs have been introduceded. Meta's Large Language Model Meta AI (LLaMA) family (Touvron et al., 2023a,b) offers models that are computationally efficient and accessible to the research community, with versions as small as 1 billion parameters. Other prominent open-source alternatives include Mistral 7B and Mixtral (Jiang et al., 2024)—a mixture-of-experts model using sparse attention to activate only a subset of its parameters during inference—and Qwen (Bai et al., 2023), a high-performing model widely adopted in academia due to its open availability and strong benchmark results.

These models showcase that decoder-only architectures when scaled massively are able to achieve broad generalization across NLP tasks with minimal supervision.

However, despite their impressive capabilities, LLMs pose significant challenges. These include high computational costs, issues related to interpretability, and the risk of generating biased or harmful content (Ferrara, 2023; George et al., 2023). While decoder-only LLMs excel in generative and open-ended tasks, encoder-only

models, such as those based on the BERT architecture, remain highly competitive and often superior for certain structured tasks, like text classification and Named Entity Recognition (NER), where their bidirectional context is advantageous. Moreover, encoder-based models tend to be more efficient to train and deploy for such applications.

2.2 Social media platforms

Social media platforms have become central to the study of natural language in real world environments, offering vast amounts of real-time, user-generated content. These platforms differ significantly from traditional text sources such as books, news articles, or formal documents. Their informal, dynamic, and highly contextual nature presents unique opportunities and challenges for NLP research and applications (Mustofa and Saptomo, 2025). A lot of these paltforms focus on different modalities, for example, Instagram emphasises photos and videos, while others like Snapchat are inherently multimodal. In contrast, Twitter and Reddit have emerged as particularly rich sources of linguistic data and are widely used in computational social science and NLP research (). This is primarily because text is the dominant medium of communication on these platforms, and they used to offer relatively easy access to data through official APIs [2]. This thesis focuses on these two platforms due to their complementary structures and modes of interaction.

Twitter is structured around short, standalone messages called "tweets," which are constrained to 280 characters. Although tweets are restricted in length, they often convey substantial information and sentiment, enhanced by the use of hashtags,

[2] Twitter's Academic API shut down on XXX. Reddit's API shutdown on XXX

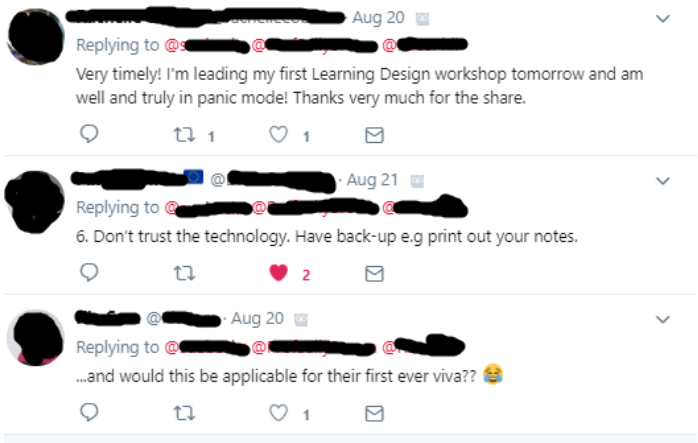

Figure 2.3: An example of a Twitter post and replies to it.

mentions, emojis, and embedded media. User interactions take the form of replies, retweets, and quote tweets, which together create loosely connected conversational threads (Figure 2.3). The character limit encourages users to be linguistically inventive, resulting in the widespread use of slang, abbreviations, and non-standard grammatical constructions.

Reddit, in contrast, is a forum-based platform organized into topic-specific communities known as *subreddits*. Posts on Reddit can be significantly longer than tweets and often initiate threaded discussions that allow for deeper, hierarchical conversations (2.4). This structure offers richer context and more coherent discourse patterns.

Although Reddit and Twitter differ in format and interaction design, they display several common linguistic characteristics as both are informal, rapidly evolving, and rely on similar engagement mechanisms such as upvotes and likes. These features shape the type and tone of user-generated content, making both platforms valuable for studying opinion formation, discourse dynamics, and online social behaviour.

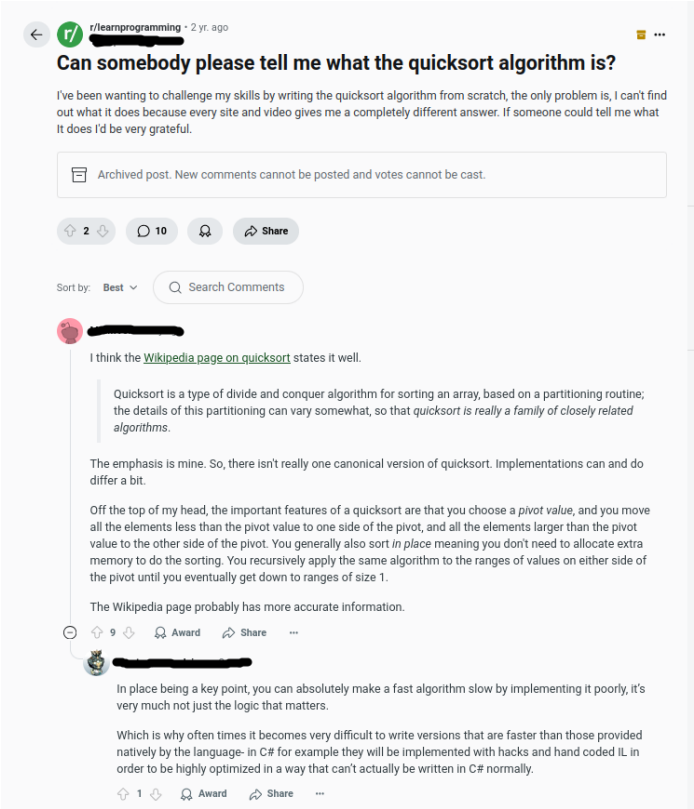

Figure 2.4: An example of a Reddit thread from r/learningprogramming subreddit.

2.2.1 Specialised Language Models for Social Media

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The direct application of LMs trained on formal or curated text corpora often fails to generalize effectively to social media data. Social media text presents a wide range of linguistic challenges for NLP due to its informal and dynamic nature. It often features slang, abbreviations (e.g., "brb", "imo"), and non-standard grammar, which deviate from conventional language norms. Posts frequently lack sufficient context, leading to ambiguity in interpretation, particularly when sarcasm or irony

is involved. Additionally, platform-specific metadata such as emojis, hashtags, and mentions play an important role in conveying meaning and identifying topics. Multilinguality and code-switching are also common, with users often blending languages or dialects within a single post. Language on these platforms evolves rapidly, with new slang and trends emerging constantly, requiring models to adapt over time. Social media content also reflects societal biases and may include toxic or harmful language, posing ethical and technical challenges. Finally, the short and often noisy nature of posts, particularly on platforms like Twitter, often requires careful pre-processing or tailored model architectures.

These challenges have led to the development of specialised language models that are pre-trained or fine-tuned on large-scale social media corpora to capture the idiosyncrasies of online discourse (Nguyen et al., 2020b).

These models are explicitly designed to handle the noise and variability inherent in social media and they often incorporate strategies for addressing spelling errors, abbreviations, emojis, and platform-specific syntax (e.g., hashtags, mentions, retweet markers). By learning from in-domain data, these models are better equipped to understand the linguistic norms, stylistic features, and emergent expressions characteristic of social platforms.

Popular examples of specialised language models include: (i) BERTweet (Nguyen et al., 2020b), a RoBERTa-based model trained exclusively on 850M English tweets; (ii) Twitter-RoBERTa (Loureiro et al., 2022b), which continues training from the RoBERTa checkpoint using a large tweet corpora; (iii) Bernice (DeLucia et al., 2022) a multilingual social media model designed to handle cross-lingual and code-switched content; and (iv) DialoGPT (Zhang et al., 2019) A conversational response generation model built by fine-tuning GPT-2 on approximately 147 million

Reddit comment exchanges. These examples reflect a common trend: RoBERTa is the backbone architecture for many social media-specific models. This is largely due to its robustness, relatively low computational requirements for large-scale pretraining, and the fact that its main limitation, a short maximum context length of 512 tokens, is not a major drawback for short-form social media text. Specialized models trained on domain-specific social media data consistently outperform their general-purpose counterparts and, in some tasks, match or surpass LLMs in downstream applications such as hate speech detection, stance classification, and misinformation detection ().

2.3 NLP Applications for Social Media

In this section, we explore core NLP tasks applied to social media data and that are significant for this thesis, highlighting their importance and the unique challenges they present. These applications include sentiment analysis, emotion detection, topic classification, hate speech detection, and misinformation analysis (Table 2.1). Each of them plays a crucial role in understanding user-generated content and extracting valuable insights from the vast amount of data available on social media platforms.

2.3.1 Sentiment Analysis

Sentiment analysis aims to determine the emotional tone or overall sentiment expressed in a piece of text. It is commonly framed as a multi-class classification task, where a model classifies a given text as negative, neutral, or positive based on

Task	Post	Label
Sentiment Analysis	Just got a promotion at work! Feeling on top of the world #Blessed	Positive
Emotion Detection	Can't believe he forgot my bday AGAIN... smh	Anger
Hate Speech Detection	Go back to where u came from. Ppl like u don't belong here	Hate Speech
Topic Classification	The new iPhone cam is insane but battery still trash lol #TechTalk	Technology

Table 2.1: Examples of social media posts across different classification tasks.

its sentiment (Mejova, 2009). Alternative approaches include aspect-based sentiment analysis, where the model assesses the sentiment of a text towards a specific target or entity (Zhang et al., 2022). Sentiment analysis can also be modeled as an ordinal regression task, where the model assigns a score on a predefined scale (e.g. 0 to 5), with 0 representing the most negative and 5 the most positive sentiment (Saad and Yang, 2019).

On social media, sentiment analysis is crucial for understanding public opinion toward brands, products, events, or political figures. For example, Matalon et al. (2021) applied sentiment analysis to examine opinion inversion in political tweets related to the Israeli-Palestinian conflict. Similarly, Rodríguez-Ibáñez et al. (2021) analyzed Twitter posts mentioning Spanish political parties to study sentiment dynamics over time. Alaparthi and Mishra (2021) used sentiment analysis to investigate public attitudes toward COVID-19 vaccines on Twitter.

Other researchers have explored the use of sentiment analysis in brand monitoring and customer feedback. Chowdhury (2024) proposed a framework for analysing customer reviews on social media to help businesses understand customer satisfaction and improve their products or services. Similarly, Chamlertwat et al.

(2012) demonstrated how sentiment analysis can be used to extract consumer insights from social media data, enabling companies to adapt their strategies based on public opinion.

Sentiment analysis has also been employed to assess public reactions to major events, such as natural disasters or political elections. For instance, Willson et al. (2021) used sentiment analysis to examine Twitter reactions to the 2019–2020 Australian bushfires, revealing regional and temporal variations in sentiment. Similarly, Behl et al. (2021) applied sentiment analysis to tweets related to the COVID-19 pandemic, helping identify user concerns and needs during the crisis and illustrating how social media can inform disaster response strategies.

In this thesis, sentiment analysis, framed as a multi-class problem, is used extensively to extract insights into users' opinions and attitudes on various topics discussed on social media. This is particularly prominent in Chapters **??**, **??**, and **??**.

2.3.2 Emotion Detection

Emotion detection while similar to sentiment analysis it is more complex and challenging task as it aims to identify more granular human emotions rather than overall negative, neutral, and positive emotional state. Specifically, an emotion detection model provided a text will highlight specific emotions like anger, joy, sadness, fear, surprise, disgust, etc that are present in it. Emotion detection can be framed as a multi-class classification task (Balabantaray et al., 2012), where each class corresponds to a specific emotion, or as a multi-label classification task (Zhang et al., 2020), where multiple emotions can be detected simultaneously in a

single text. It can also be approached as an ordinal regression task, where the model assigns a score to each emotion (e.g. 0-5) based on its intensity or relevance to the text (Muhammad et al., 2025).

Emotion detection is important for social media analysis particularly relevant when there is need for more fine-grained understanding of emotional responses to events or content on social media. Researchers have utilised emotion detection to study political discourse, marketing campaigns, and hate speech studies among others. For example Bhaumik et al. (2023) apply emotion detection techniques on Twitter posts related to the 2017 French presidential elections and study the effect of emotional content as influence indicators in online political discussions.Researchers (Rahib et al., 2022), have also used emotion detection to explore the effect of the COVID-19 pandemic to the general population and study how they are responding to and dealing with evolving circumstances. Finally, Schäfer and Kistner (2023) investigate the connection between hate speech and emotions and propose a method for jointly modeling both phenomena.

In this thesis, emotion detection, framed as multi-label task, is utilised in Chapters **??**, and **??** as part of our analysis to extract insights from users about the emotional status.

2.3.3 Topic Modelling

Topic modelling is focuses on identifying the underlying themes or topics present in a collection of texts. It is often framed as an unsupervised learning task, where the model discovers latent topics without predefined labels. Common approaches include Latent Dirichlet Allocation (LDA) (Blei et al., 2003), Non-negative Matrix

Factorization (NMF) (Lee and Seung, 1999), and more recent methods like BERT-based topic models such as BertTopic (Grootendorst, 2022) and Top2Vec (Angelov, 2020). These methods aim to group similar documents based on their content, allowing for the identification of prevalent themes within a corpus.

The unsupervised nature of topic modelling while beneficial in many cases, can also be a limitation when specific topics or categories are of interest. In such cases, supervised topic classification is often used, where a model is trained on labeled data to predict predefined topics for new texts. This approach is particularly useful when performing targeted analysis where specific specific categories, such as politics, technology, and sports are of interest. This is particular important on social media, as it helps in organizing vast amounts of information, identifying trending discussions that are of interest and filter content based on specific themes.Applications of topic classification on social media focus on identifying trending topics (Lee et al., 2011; Bouadjenek et al., 2022) but have also been utilised often in parallel with sentiment analysis to gain insights into public opinion and discourse (Huang et al., 2013; Rodrigues and Chiplunkar, 2022).

In this thesis we utilise topic classification in Chapters **??**, **??**,**??** with the aim of community profiling and identifying topics of interest for specific groups of users.

2.3.4 Hate Speech Detection

Hate speech detection involves identifying offensive, threatening, or discriminatory language. It is a critical application of NLP on social media, aiming to identify content that expresses hatred or encourages violence towards individuals or groups based on attributes like race, ethnicity, religion, gender, and sexual orientation.

This is a highly sensitive and challenging task that is crucial for moderation and creating safer online environments. Hate speech detection can be framed as a binary classification task, where the model predicts whether a given text contains hate speech or not (Saleh et al., 2023) but can an also be approached as a multi-class classification task where the model identifies specific types of hate speech (e.g., racism, sexism, homophobia) (Qureshi and Sabih, 2021), or as a multi-label classification task, where multiple forms of hate speech can be present in a single text (Dirting et al., 2022).

Hate speech detection is particularly important on social media platforms, where the rapid spread of harmful content can have significant real-world consequences as was the case during the Rohingya crisis in Myanmar (Nourooz Pour, 2023). Researchers have applied hate speech detection to various social media platforms, including Twitter and Reddit, to study the prevalence and impact of hate speech in online discourse. For example, Kim and Kesari (2021) investigates relations between misinformation and hate speech specifically in the case of the COVID-19 pandemic and hate speech target to Asian communities. Other researchers (Agarwal et al., 2021) have explore the nature and prevalence of hate speech in political dialogue in Twitter and also sutdies that focus on specific expressions of hate speech like misogyny and abuse of female politicians on Twitter (Fuchs and Schäfer, 2021).

In this thesis, hate speech detection is framed as a multi-class classification task and is used in Chapters **??**, **??**, and **??** to identify harmful content and understand the dynamics of online hate speech.

2.3.5 Misinformation & Polarisation

WIP

Misinformation and disinformation analysis is a critical area of NLP research, particularly in the context of social media platforms where the rapid spread of false or misleading information can have significant societal impacts. The spread of misinformation on social media contributes to public harm, especially in contexts like health communication and political discourse. Identifying false or misleading content is complicated by subtle rhetorical strategies and the rapid evolution of narratives.

Misinformation analysis includes tasks such as misinformation detection, stance detection, propaganda detection, source credibility assessment, and linguistic feature analysis. These tasks are essential for understanding how misinformation spreads, its impact on public opinion, and how to counteract it through fact-checking and verification efforts. Misinformation detection involves identifying false or misleading content, often framed as a binary classification task where the model predicts whether a given text contains misinformation or not (). Stance detection, on the other hand, determines whether a text agrees, disagrees, discusses, or is unrelated to a given claim, and can be framed as a multi-class classification task (). Propaganda detection focuses on identifying rhetorical techniques used to manipulate public opinion, often framed as a multi-class classification task where the model identifies specific propaganda techniques used in a text (). Source credibility assessment involves evaluating the reliability of the source of information, which can be framed as a binary classification task where the model predicts whether a source is credible or not (). Finally, linguistic feature analysis involves

analyzing linguistic patterns (e.g., sensationalist language, emotional intensity, specific vocabulary) often associated with misinformation, which can be framed as a feature extraction task where the model identifies specific linguistic features in a text ().

On the other hand polarisation analysis focuses on understanding the strong division of the populace in two or more sides (), and the formation of echo chambers and selective exposure, where users are exposed only to information that aligns with their existing beliefs. Polarisation often leads to the reinforcement of biases, the spread of misinformation, and the deterioration of constructive discourse (). Tasks related to polarisation can include echo chamber detection, opinion dynamics tracking, stance and ideology detection, linguistic markers of polarisation, and studying disagreement and deliberation in online discussions. Echo Chambers and Filter Bubbles: NLP can identify communities with shared beliefs and analyze how information flows within and between these groups. Opinion Dynamics: Tracking how opinions evolve over time and how they are influenced by social interactions and exposure to different narratives. Linguistic Markers of Polarisation: Analyzing specific language choices (e.g., divisive rhetoric, "us vs. them" framing, emotional language) that indicate or contribute to polarization. Disagreement and Deliberation: Studying how online discussions devolve into unproductive arguments versus constructive dialogue.

In this thesis we look at both misinformation and polarisation effects, specifically in Chapter **??** by investigating user profiles likely to share misinformation, and in Chapters **??** **??** focuses on polarisation in social discourse.

Part I

Modeling Approaches for Social Media Language Understanding

3

Twitter Topic Classification

Social media platforms host discussions about a wide variety of topics that arise everyday. Making sense of all the content and organising it into categories is an arduous task. A common way to deal with this issue is relying on topic modeling, but topics discovered using this technique are difficult to interpret and can differ from corpus to corpus. In this chapter, we present a new task based on tweet topic classification and release two associated datasets[1][2]. Given a wide range of topics covering the most important discussion points in social media, we provide training and testing data from recent time periods that can be used to evaluate tweet classification models. Moreover, we perform a quantitative evaluation and analysis of current general- and domain-specific language models on the task, which provide more insights on the challenges and nature of the task.

[1] https://huggingface.co/datasets/cardiffnlp/tweet_topic_single
[2] https://huggingface.co/datasets/cardiffnlp/tweet_topic_multi

3.1 Introduction

Social media platforms, e.g., *Twitter, Snapchat, TikTok and Instagram*, provide an environment for content creation and information sharing among people. On social platforms, every individual can express their views about current events or anything that they care about, influencing and guiding discussions among their friends and followers. We can think about the 2020 Black Lives Matter protests where in a period of 12 days more than 47.8 million posts mentioned the hashtag #BlackLivesMatter on *Twitter* (Anderson et al., 2020) or the UK's 2016 Brexit referendum where more than 200 million relevant tweets had been shared in 3 years (June 2016 -January 2019) (National, 2019).

Social media platforms are highly studied to understand behaviours among users, groups, organizations, or even societies (Yang et al., 2021), and in particular to understand opinion of people regarding a variety of topics such as politics (Zhuravskaya et al., 2020), diversity and inclusion (Chakravarthi, 2020), TV shows (Wohn and Na, 2011), sports events (Lim et al., 2015), or finance (Hu et al., 2021). However, one of the biggest challenges in understanding this type of user generated content, is the noise and variety of these texts (Morgan and Van Keulen, 2014; Baldwin et al., 2013). Consequently, identifying topics within social media platforms from their posts is not a trivial task.

Existing solutions can be divided into topic modeling and topic classification. For topic modeling, topics are detected in an unsupervised way with models such as LDA (Blei et al., 2003) and subsequent variations (Steyvers and Griffiths, 2007). Similarly, solutions that use new BERT contextualized embeddings (like BERTopic

(Grootendorst, 2022)) have increased in popularity as they offer increased performance. However, these approaches assume that (i) all the topics of interest are represented in the documents included in the study, and (ii) the terms present in these documents are enough to characterize each topic. For these reasons, these methods are usually built as an ad-hoc analysis. Another limitation of these models is interpretability, as it is hard to generalize and label each cluster topic.

On the other hand, topic classification approaches the problem in a supervised manner and assigns multiple topics to each document based on a predefined set of categories. This approach overcomes the issues of interpretability and is not based on assumptions about the vocabulary distribution mentioned above. However, the downside of topic classification is that relies on curated datasets labeled by human annotators, and this can be expensive and time consuming to create.

In this chapter, we introduce {TweetTopic}, a topic classification dataset on *Twitter* data. To the best of our knowledge, this is the first large-scale topic classification dataset specifically tailored to social media, rather than standard text as news articles (Greene and Cunningham, 2006) or scientific papers (Lazaridou et al., 2021).

The dataset consists of a total of 11,267 tweets collected through a time period from September 2019 to August 2021. Each tweet is assigned one or more topics from a predefined set of categories curated by social platform experts. Aiming to test the robustness of our dataset through time and across topics, we perform several classification experiments, both single-label and multi-label, while utilizing state-of-the-art language models.[3]

[3]Tweet classification models associated with TweetTopicChave been integrated into TweetNLP (Camacho-Collados et al., 2022).

3.2 Related Work

3.2.1 Social media.

Social media have become an important aspect of the daily life of millions of people, with 81% of adults in the U.S. stating to have used at least one social platform in 2021 (Auxier and Anderson, 2021) and over 57% of people in EU interacting through social media in 2020 (Eurostat, 2021). In recent years, an increasing number of corporations seem to dedicate a more significant portion of their marketing funds to advertising on social platforms compared to other more traditional mediums (Eid et al., 2020). At the same time, social media has become a political battleground where politicians both debate between them and try to communicate with their voters, (Stier et al., 2018; Llewellyn and Cram, 2016). Finally, social platforms have been used extensively by their users as a means for almost instantaneous news updates both for day-to-day events (Hermida, 2012), and human and natural disasters (e.g., the Ukrainian war or the COVID-19 pandemic) (Khaldarova and Pantti, 2016; Banda et al., 2021).

Therefore, a large volume of content is being generated in social media everyday. Its polymorphism also means that performing any targeted analysis on the data can be a challenging and time-consuming process (Weller, 2015; Stieglitz et al., 2018). Furthermore, even though there are various existing tools focused on analyzing social media data (Batrinca and Treleaven, 2015), there is no established way to efficiently identify and filter only relevant and valuable content (Nugroho et al., 2020).

3.2.2 Topic modeling.

Topic models are unsupervised methods to identify relevant topics given a text corpus. LDA (Blei et al., 2003) is one of the most popular algorithms for topic modeling. However, despite being successful in identifying topics in traditional media (Martin and Johnson, 2015; El Akrouchi et al., 2021), LDA often struggles when applied to short, unstructured, and constantly evolving texts, such as *Twitter* data (Zhao et al., 2011). It also typically underperforms when compared to other supervised methods (Arias et al., 2015). More recently, several variations of LDA have been proposed to address these challenges with social media texts, such as combining author-topic modelling with LDA (Rosen-Zvi et al., 2004; Steinskog et al., 2017), frameworks like Twitter-LDA (Zhao et al., 2011) where noisy words and author information are taken into account, and SKLDA (Tajbakhsh and Bagherzadeh, 2019), where semantic relations between words extracted from WordNet are taken into account.

However, LDA-based methods are often not ideal when we need to assign more than one topic to a document. Even though there are approaches to acquire multiple labels for each topic, they are usually based on hierarchical (Griffiths et al., 2003) or graph (Li and McCallum, 2006) architectures which, depending on the use case, make assumptions about relations of the topics that may not be present in a given corpus (i.e. parent/children topics). Furthermore, semi-supervised or supervised variations of LDA, such as PLDA (Ramage et al., 2011) and sLDA (Mcauliffe and Blei, 2007), have been been used on *Twitter* data (Resnik et al., 2015; Ashktorab et al., 2014). While such methods have potential for increased performance they usually require prior labelling or information about the documents and thus remove a major advantage they have compared to supervised

approaches.

Finally, as a mainly unsupervised technique, evaluating the results of topic modeling can be a hard task. Metrics such as purity, mutual information and pairwise F-measure are used to evaluate the quality of topics/clusters created by the models (Nugroho et al., 2020). On the other hand, qualitative analysis is usually difficult to perform due to the lack of interpretability of topics produced and the difficulty increases with the amount of topic.

In contrast to traditional LDA approaches, techniques such as BERTopic (Grootendorst, 2022) and Top2Vec (Angelov, 2020) attempt to make use of existing knowledge from pretrained language models by extracting embedding representations of tweets and using them to perform topic clustering. Both BERTtopic and Top2Vec tend to be easier to use than LDA, without the need for extensive hyper-parameter tuning, and often result in increased performance (Egger and Yu, 2022). However, they do have disadvantages, namely: not performing well on small datasets (Abuzayed and Al-Khalifa, 2021), generating a lot of outlier topics (Silveira et al., 2021), and requiring existing knowledge. Finally, these approaches suffer similar drawbacks to LDA regarding evaluation and interpretability.

3.2.3 Topic classification.

Given a text as an input, topic classification is the task of associating it with a specific topic (or topics) from a pre-defined set of categories. In what concerns social media, previous work has focused on predicting hashtags as classes (Dhingra et al., 2016). However, the dynamic nature of the events discussed in those

platforms makes any dataset focused on hashtags quickly become sparse and outdated. Any new model needs to be trained from scratch since the category set will be different based on the relevance of hashtags. Nevertheless, by focusing on higher-level topics like *Sports* or *Arts & Culture*, widespread and recurrent in social platforms, the data can be leveraged for more extended periods, and any model trained on it can be easily updated with more data as the label set is fixed. It also improves interpretability since there is a clear semantic meaning to the proposed categories, while hashtags might be ambiguous or require additional interpretation.

In terms of previously released data, existing datasets mainly focus on the news articles domain, e.g., BBC News (Greene and Cunningham, 2006), Reuter (Lewis et al., 2004), 20 Newsgroups (Lang, 1995), and WMT News Crawl (Lazaridou et al., 2021) with few exceptions like scientific (arXiv) (Lazaridou et al., 2021) and medical (Ohsumed) (Hersh et al., 1994) domains. Therefore, these datasets offer different sets of challenges with respect to social media.

3.3 Tweet Topic Classification

This section presents the pipeline to construct TweetTopic, our topic classification dataset based on *Twitter* data. This pipeline is divided into three steps: (i) tweet collection, (ii) data filtering, and (iii) topic annotation. These steps are explained in more detail in the next subsections.

3.3.1 Tweet collection

Our goal is to collect a set of tweets with a high coverage of diverse topics over time. We fetched the tweets given specific keywords and time periods using the *Twitter* API. Since the tweets returned by the API are in reverse chronological order, we decided to split the queries into small time windows to make sure that the tweets are distributed over time. In our case, we queried 50 tweets every two hours from September 2019 to October 2021. As the keywords used to create queries, we collected lists of trending topics from *Snapchat*[4] in each week during the period (e.g. *pink super moon*, *social distancing*, and *NBA*).

This step allowed us to collect tweets with a similar distribution to topics in the real world over time. For this step we also added conditions to exclude retweets, replies, quotes, and tweets with media, as well as specifying the language as English only. In the end, we collected a total of 1,264,037 raw tweets from the API.

3.3.2 Data Filtering

Tweet filtering. Since the raw tweets may contain irrelevant content, we applied several text filtering techniques to get a cleaner tweets corpus. Our text filtering pipeline consists of two steps as described in Figure 3.1: *pre-filtering* and *near-deduplication*. This filtering fulfilled different goals such as removing abusive content, improving quality and avoiding near-duplicates. In the *pre-filtering*, we first removed non-English tweets by using a fastText based language identifier[5]

[4]Available at https://trends.snapchat.com/. We were not able to access *Twitter* trends since they are not publicly available through APIs.

[5]https://fasttext.cc/blog/2017/10/02/blog-post.html

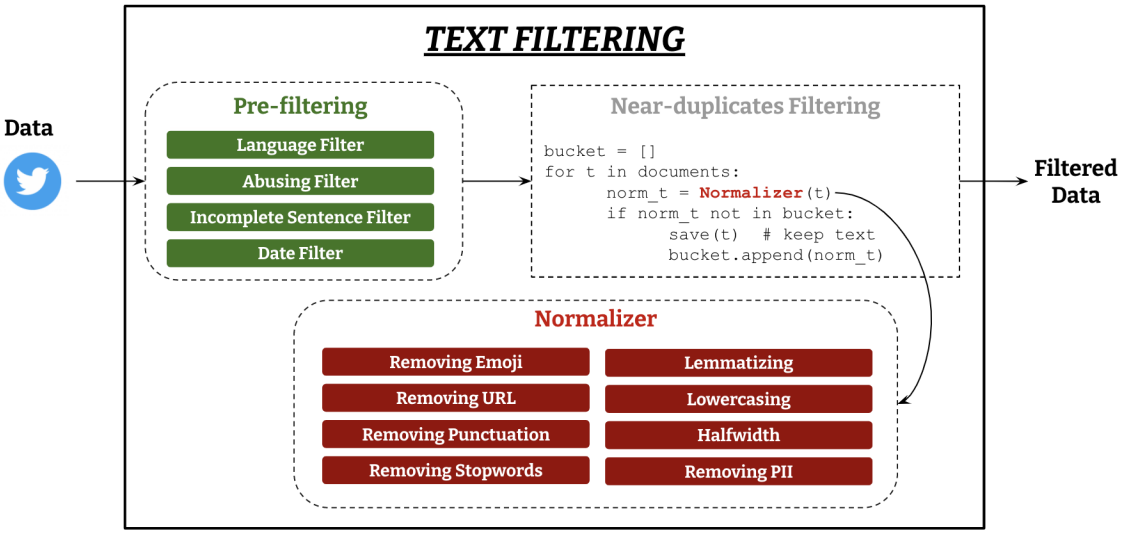

Figure 3.1: Text filtering pipeline to reduce noise from the tweets and avoid near duplicates.

(Bojanowski et al., 2016). Then, we removed tweets that contained incomplete sentences (e.g., too short or end in the middle of the sentence) or abusing words by using rule-based heuristics. Then, we applied a *near-duplication* filter to drop duplicated tweets. In particular, we first normalized each tweet, and kept unique tweets only in terms of their normalized form. The normalizer first converted full-width to half-width and removed substrings from the tweet such as emoji, web URLs, punctuation, stopwords, and personally identifiable information (PII).[6] Then, we lemmatized and lowercased each word in the tweets and removed identical tweets after normalization.

Trend filtering. Given our budget and in order to further reduce the number of tweets to annotate while ensuring diversity, we grouped the tweets by the trending topics used to query the raw tweets in each week, and selected the top 15

[6]We detected PII with scrubadub and other components are all based on NLTK.

most common trends within the week.[7] We applied the trending topic filtering for every week which resulted in our final dataset, consisting of 28,573 tweets in total. Note that the trends are different every week, so the tweets are diverse across weeks regarding the trends.[8]

3.3.3 Annotation

To attain topic annotations over the tweets, we conducted a manual annotation on Amazon Mechanical Turk. We randomly sampled 11,374 tweets from the cleaned tweets and each tweet was annotated by five annotators, collecting 56,870 annotations in total. We manually constructed a topic taxonomy that contained 23 initial topics across diverse genres, asked workers to annotate the relevant (possibly multiple) topics to the tweet.[9] The initial list of 23 topics was shared with us by a research team of Snapchat. This list was selected and curated by a team of social media experts from the company over time to ensure a tailored coverage of social media content.

We ensured several quality control mechanisms within the test, including a qualification test. Each tweet was annotated by five turkers and the final budget for the total estimated annotation cost was $4,000. Each single assignment contained 50 tweets to be annotated where each annotation is completed with an interface that we include in the Appendix **??**.

[7] More details about this process can be found in the Appendix **??**.

[8] In the Appendix **??** we provide a detailed breakdown of the distribution of trends in each week. There, we can confirm that the top trend does not go beyond 20% in most cases, which ensures a diverse set of trends.

[9] The actual instructions shown to workers are included in the Appendix **??**.

Raw	Pre-filter	De-duplication	Trend-filter	Annotated
1,264,037	596,028	202,604	28,573	11,267

Table 3.1: Number of total tweets after each step.

As quality control, each assignment contained three qualification tweets and only those who annotated them correctly were accepted. A small number of raters (10) and their respective tweets were also discarded as they displayed unusual behaviour selecting on average more than 5 labels for each tweet where the global average was 1.6 labels per tweet. Also, workers were not allowed to work on the assignment more than once.

Post-aggregation. We followed Mohammad et al. (2018) by assigning a label to a tweet provided that the label was suggested by at least two annotators. We opted out of a majority rule as this way our dataset can be used to develop more robust systems that can handle real-world data, which are rarely straightforward and instead can often contain complex linguistic phenomena (Mohammad et al., 2018). Tweets where none of the classes received at least two votes were discarded. The number of tweets in each process is summarized in Table 3.1.

Inter-annotator agreement. Several metrics can be used to evaluate the quality of an annotation task (Artstein and Poesio, 2008) and it is often difficult to select the most appropriate one. In our experiment, we utilized Krippendorff's alpha (Krippendorff, 2011) with MASI distance (Passonneau, 2006), which is a common combination when dealing with multi-rater and multi-label tasks (Artstein and Poesio, 2008). For our task the alpha statistic results in 0.35. As a comparison reference, a completely random annotation would produce a 0 alpha statistic.

Tweet	Topics
Apple Removed More Than 30,000 Apps From The Chinese App Store	- business & entrepreneurs - news & social concern - science & technology
#copreps Football: End of the line for FLHS season	sports & games

Table 3.2: Sample tweets for each setting studied (top: multi-label; bottom: single-label).

When considering the percent agreement of each pair of annotators we acquire a value of 0.87 in contrast to 0.62 for random annotation. These inter-annotator agreement results appear to be inline or slightly better then previous similar multi-label annotation tasks (Mohammad et al., 2018).

3.3.4 Settings and temporal split

In order to investigate potential temporal differences in the corpus we split the datasets into two periods: (1) from September 2019 to August 2020 (referred to as training data) and (2) from September 2020 to August 2021 (test data). The motivation behind this temporal split is to make the task more realistic and evaluate the generalizability performance of the classifiers on future data.

We established two classification settings: (1) multi-label and (2) single-label. Sample instances from both settings are displayed in Table 3.2.[10] With this distinction, we aim to provide flexibility to users, and increase the usability of the dataset for settings and analyses, where a more fine-grained classification of tweets is not required (i.e. single-label).

[10]For readability, tweet examples have been slightly modified, removing links and usernames which are anonymized in the dataset.

Multi-label. By applying a final post-aggregation step to exclude categories that may not be relevant for social media, we removed those categories with fewer than 50 labels overall, leaving a final set of 19 topics.

Single-label. In an effort to keep the classes relatively balanced, we firstly excluded tweets that were labeled with the most dominant of the classes, i.e., news & social concern (32.82% of total tweets), which is highly cross-category. Following this, the remaining ten most prominent classes were considered. Finally, based on logical assumptions regarding the similarity of the classes and also the overlap between them, several labels were grouped together. More specifically: *gaming* and *sports* (35% overlap) were grouped as *sports & gaming*; *music*, *celebrity & pop culture*, and *film tv & video* (44% and 31% overlap) became *pop culture*; *diaries & daily life* and *family* (54% overlap) were grouped together as *daily life*. These three new classes along with the original *arts & culture*, *business & entrepreneurs*, and *science & technology* composed the final set of topics. Finally, in this setting, tweets containing more than one of these six labels were dropped.

3.3.5 Statistics

The final set of annotated tweets is 11,267 and 6,997 for the multi-label and single-label settings, respectively. Figures 3.2 and 3.3 display the percentage of tweets that were classified in each topic, for each time period studied, after the aggregation of annotations for multi-label and single-label settings, respectively.[11] The imbalanced nature that can be observed, e.g., *sports* consisting of 26% of the

[11] For the multi-label setting the percentages sum up to more than 100% due to the nature of the annotation.

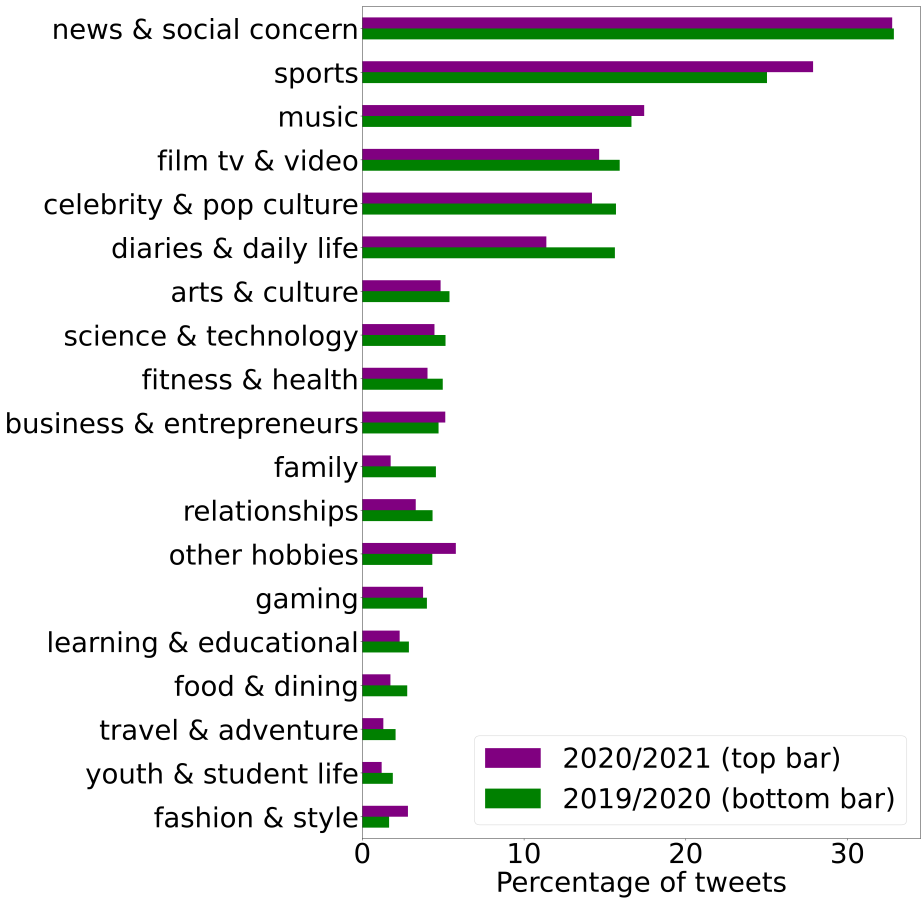

Figure 3.2: Percentage of tweets that were annotated with a given topic (multi-label setting) for each time period.

2019/20 multi-label dataset while *travel & adventure* only 2%, is explained due to the way tweets were collected, where we aimed to mimic the distribution of real-world data on *Twitter*.

Number of labels. When considering the multi-label setting, 50% of the tweets are classified with only one label while only 2.7% are given four or more labels, with the maximum amount being six. However, the dataset is diverse enough to with 35% and 12% of the tweets having two and three labels respectively. This

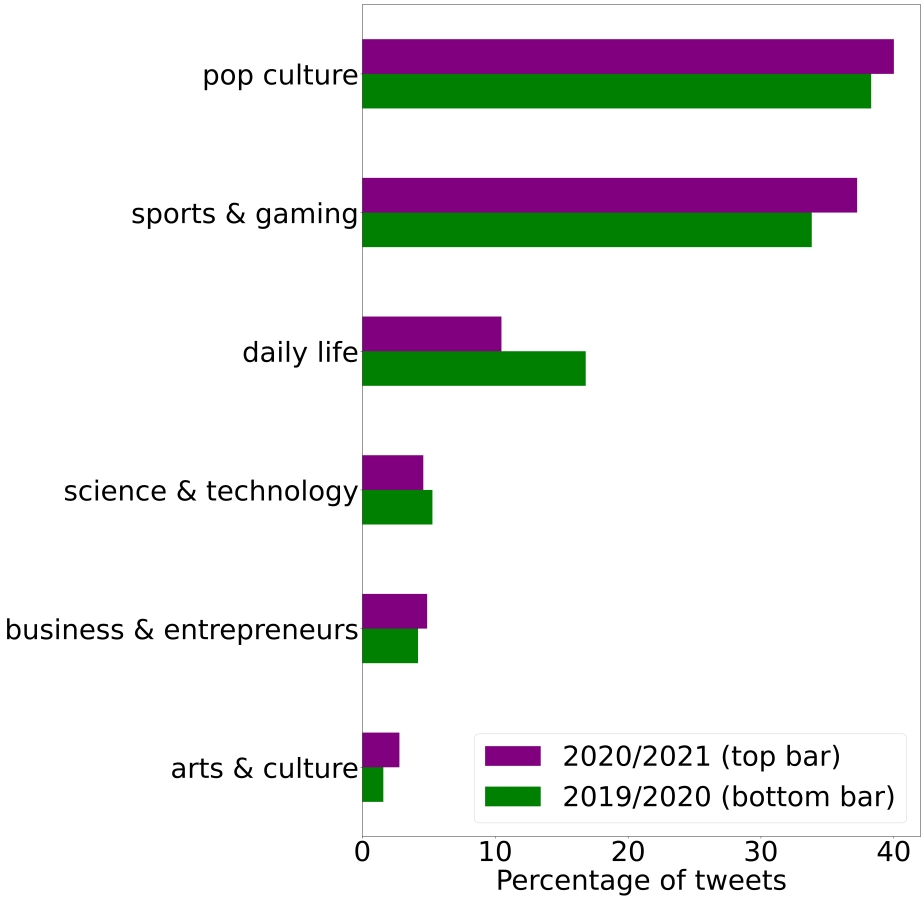

Figure 3.3: Percentage of tweets that were annotated with a given topic (single-label setting) for each time period.

coder behaviour (i.e. preferring to select only one class) can be observed on similar multi-label annotation tasks (Véronis, 1998; Poesio and Artstein, 2005).

Class distribution across time periods. We note that the distribution of classes between the two time periods studied remains largely similar in both settings with the largest difference being in the *music* and *news & social concern* classes being 3.5% more populous in 2019/20. This observation suggests that our curated topics are broad enough to be relatively robust to temporal trends.

Class	length	punc	upp/low	#	@	emojis	mtld	count
arts & culture	166.9 ±67.5	6.5 ±3.4	0.2 ±0.6	0.8 ±1.4	0.4 ±0.5	0.1 ±0.3	140.9	577
business & entrepreneurs	186.3 ±65.5	6.4 ±3.1	0.1 ±0.2	0.6 ±1.1	0.5 ±0.5	0.0 ±0.2	159.0	554
celebrity & pop culture	155.5 ±67.8	7.4 ±3.7	0.2 ±0.9	0.6 ±1.0	0.8 ±0.7	0.1 ±0.4	145.8	1685
diaries & daily life	168.3 ±68.4	5.4 ±3.3	0.1 ±0.7	0.4 ±0.9	0.4 ±0.5	0.1 ±0.5	132.5	1525
family	165.1 ±68.5	5.2 ±3.2	0.2 ±1.4	0.5 ±1.0	0.4 ±0.5	0.2 ±0.5	112.7	358
fashion & style	147.9 ±55.4	7.8 ±3.1	0.2 ±0.5	1.0 ±1.5	0.6 ±0.5	0.1 ±0.3	98.8	251
film tv & video	157.7 ±66.3	7.5 ±3.7	0.2 ±0.8	0.6 ±1.1	0.7 ±0.6	0.1 ±0.4	145.1	1723
fitness & health	195.4 ±67.1	6.3 ±2.8	0.1 ±0.1	0.5 ±0.9	0.6 ±0.5	0.1 ±0.3	168.5	508
food & dining	165.2 ±64.5	6.1 ±3.1	0.1 ±0.2	0.5 ±1.0	0.4 ±0.5	0.1 ±0.4	154.7	255
gaming	159.6 ±68.9	6.5 ±3.9	0.1 ±0.2	0.5 ±1.0	0.5 ±0.6	0.0 ±0.2	128.4	437
learning & educational	191.8 ±65.8	5.9 ±2.9	0.1 ±0.1	0.6 ±1.0	0.5 ±0.6	0.0 ±0.2	156.7	293
music	143.5 ±64.0	8.4 ±4.4	0.3 ±1.1	0.7 ±1.1	0.8 ±0.7	0.1 ±0.5	119.8	1919
news & social concern	183.1 ±70.5	6.6 ±3.0	0.2 ±1.3	0.4 ±0.8	0.6 ±0.6	0.0 ±0.2	165.1	3698
other hobbies	160.9 ±69.2	6.3 ±3.4	0.2 ±0.7	0.6 ±1.0	0.4 ±0.6	0.1 ±0.4	143.6	568
relationships	162.4 ±70.6	5.3 ±3.5	0.2 ±1.6	0.4 ±0.9	0.5 ±0.6	0.2 ±0.9	111.9	432
science & technology	177.9 ±69.4	6.7 ±2.8	0.1 ±0.5	0.5 ±1.0	0.6 ±0.5	0.0 ±0.1	164.2	542
sports	162.8 ±65.9	6.4 ±3.2	0.2 ±1.4	0.5 ±0.8	0.7 ±0.6	0.1 ±0.3	152.8	2977
travel & adventure	175.2 ±72.3	6.2 ±3.1	0.2 ±1.8	0.5 ±1.0	0.5 ±0.5	0.1 ±0.2	173.1	190
youth & student life	202.0 ±62.4	5.9 ±3.2	0.1 ±0.1	0.5 ±0.9	0.5 ±0.6	0.1 ±0.2	155.6	174

Table 3.3: General lexical statistics for each class. The averages of the length of tweet, punctuation count, upper/lower case ratio (upl/low), hashtags count, mentions count, emojis count are reported along with their standard deviation. Frequency metrics are normalized based on the text length. The last two columns correspond to the lexical diversity (mtld) and total number of tweets.

Topic features. In order to get a better understanding of the data, and to investigate potential significant characteristics, we extract various statistics from the tweets in the multi-label dataset. Table 3.3 displays the average values of tweet length, number of punctuation symbols, upper to lower case ratio, number of hashtags, number of mentions and number of emojis, along with their standard deviations for each topic. In order to have a fair comparison, all the metrics are normalized based on the tweet length ($(metric/length) * 100$). The Measure of Textual Lexical Diversity (MTLD) (McCarthy and Jarvis, 2010) is also reported as an indication on the vocabulary richness of each class, as well as the number of tweets for each class.

The topics *celebrity & pop culture* and *music* have the highest occurrences of

mentions "@" (0.8). This is intuitively due to the fact that a large number of tweets belonging to these classes will mention recognizable users such as artists or athletes. Similarly, tweets belonging to the *fashion & style* topic tend to include more hashtags (#) on average (1 hashtag per tweet), which can be attributed to the nature of hashtags in *Twitter*, usually employed to indicate popular and trending topics. Finally, topics that can be considered more accessible to the general public such as *fashion & style*, *family*, and *relationships* achieve a relatively low lexical diversity score (98.8, 112.7, 111.9) while more specialized or advanced topics such as *travel & adventure*, *business & entrepreneurs* and *fitness & health* display higher lexical diversity (173.1, 159.0, 168.5).

3.4 Evaluation

3.4.1 Experimental setting

Datasets. We perform experiments in our tweet classification annotated datasets. In particular, our experiments are based on two settings, single-label and multi-label (see Section 3.3.4 for details).

Comparison systems. To evaluate our dataset, we first use simple baselines: Majority (most frequent class in training) and Random (uniform probability for each class). As comparison systems, we train a traditional bag of words with SVM and a fastText classifier (Bojanowski et al., 2016) that utilizes pretrained embeddings (Mikolov et al., 2018). Furthermore, BERT base and large (Devlin et al., 2018) and both base and large versions of RoBERTa (Liu et al., 2019b) are

used as comparison systems. As classifiers specialized on social media, i.e. trained on *Twitter* data, BERTweet (Nguyen et al., 2020a), TimeLM-19, and TimeLM-21 (Loureiro et al., 2022a), all based on a RoBERTa architecture, are also utilized. BERTweet is trained on a corpus of 845M tweets mainly from 01/2012 to 08/2019, while also including 5M COVID-19 related tweets from 01/2020 to 03/2020. On the other hand, TimeLM-19 is trained on 95M tweets gathered between 2018 and 2019. For completeness, we also report results of TimeLM-21, trained on 125M tweets from 2018 to 2021, but excluded it from our main analysis given the time overlap with the test set (reminder that one of the motivations of this task is to be able to process tweets in real time). TimeLMs models use the RoBERTa-base model as initial checkpoint, while BERTweet is trained from scratch.

The implementations provided by Hugging Face (Wolf et al., 2019) are used to train and test all language models.[12]

Evaluation metrics. For both settings macro average Precision, Recall and F1, as well as Accuracy, are used to evaluate the models tested. As an alternative metric for the multi-label setting, Jaccard Index (JI) is also utilized, as it can offer useful insights about the models performances (Pereira et al., 2018; Tsoumakas et al., 2009). More specifically, the index is calculated for each tweet individually and the final metric is computed as the average over all entries.

3.4.2 Results

Table 3.4 displays the results of all comparison system on both settings. While only a number of models were tested, the results suggest that domain-specific knowledge

[12]More details about the exact hyperparameters are included in the Appendix **??**.

	Model	Multi-label					Single-label			
		Pr	**Rec**	**F1**	**Acc**	**JI**	**Pr**	**Rec**	**F1**	**Acc**
Baselines	Random	8.4	48.3	12.6	0	7.9	15	14.2	11.9	15.5
	Majority	1.5	5.3	2.3	18.0	22.6	6.7	16.7	9.5	40
	SVM	69.4	23.7	30.5	37.1	51.8	73.6	47.4	50.2	75.8
	fastText	67.0	18.0	24.0	31.9	43.5	56.0	46.0	48.0	74.0
Language models	BERT-base	69.7	42.5	50.1	45.5	63.9	62.4	60.0	58.8	81
	BERT-large	64.4	**51.5**	56.4	44.6	65.1	62.4	61.7	61.7	84.3
	RB-base	68.5	49.2	55.8	46.5	66.2	64.8	66.7	65.6	85.9
	RB-large	**72.2**	48.9	56.3	**47.9**	**67.7**	66.1	56.2	58.3	84.5
	BERTweet	66.9	46.1	52.7	47.1	66.9	64.9	65.6	63.8	85.2
	TimeLM-19	71.1	50.4	**57.2**	47.7	67.5	**76.5**	**68.9**	**70.0**	**86.4**
	TimeLM-21	*66.1*	*54.2*	*58.8*	*47.1*	*67.6*	*73.9*	*69.8*	*70.1*	*86.8*

Table 3.4: Macro average Precision (Pr), Recall (Rec), F1, and accuracy results in TweetTopic(temporal split). Jaccard Index (JI) is reported for the multi-label setting.

appears to be more important than the size of the model, with *Twitter* base models outperforming large generic language models. Given the larger number of labels and more challenging setting, multi-label classification appears to be most challenging setting with the best model TimeLM-21, barely achieving 58.8% F1 and 67.6% Jaccard scores, in comparison to 70.1% F1 and 86.8% Accuracy in the single-label setting. However, it is important to note that TimeLM-21 has the unfair advantage of being trained with a more recent corpus and more specifically a corpus from the same time period as the test set. Taking this into consideration, the next best performing model is TimeLM-19 with 57.2% and 70% F1 scores, for the multi-label and single-label settings respectively. Even though the differences in the average F1 scores between the two models is relatively small, 1.6% and 0.1% for multi/single settings, when taking into account their performance in each individually topic, we can identify topics where TimeLM-21 clearly outperforms TimeLM-19 (see Section 3.5.1 for more details).

3.5 Analysis

In this section, we analyse two important aspects of the TweetTopicdataset, mainly its temporal dimension (Section 3.5.1) and the errors made by the systems (Section 3.5.2).

3.5.1 Temporal analysis

The strong performance of TimeLM-21 provided evidence regarding the importance of an up-to-date training corpus. We continue our investigation by training the same set of models on a random split of the data (i.e., both training and test sets with tweets from 2019 to 2021). To make the results comparable, we created training and test sizes of the same size as the original temporal split.[13]

Table 3.5 displays the F1 scores, while using a multi-label setting for each class in both the temporal and random splits. Every model tested performs better when trained using information from both time periods, i.e using random split. Taking into account that in both splits the distribution of classes is similar (Figure 3.2), we can assume that the temporal differences in the data provide useful information. It is worth noting that the "specialized" *Twitter* models display a more robust performance regarding the training data used. In particular, there are 8, 9 and 4 topics where BERTweet, TimeLM-19, and TimeLM-21 respectively perform better while using the temporal split in contrast to 3 and 1 of RoBERTa base and large respectively (models that have a similar architecture).

[13]While the distribution of labels may naturally be altered, this change is minimal, as we can recall from Figures 3.2 and 3.3.

Class	Random		SVM		BERT		RB		RB-large		BERTweet		TimeLM-19		TimeLM-21	
	temp	rand	temp	rand	temp	rand	temp	rand	temp	rand	temp	rand	temp	rand	temp	rand
arts & culture	**8.6**	8.3	3.6	**27.7**	17.8	**35.9**	20.9	**41.2**	28.0	**44.0**	9.8	**28.2**	21.3	**39.1**	35.4	**44.8**
business & entrepreneurs	**8.7**	7.4	18.0	**28.7**	**53.3**	49.2	56.7	**57.1**	50.9	56.5	**59.7**	54.5	**58.6**	55.3	**56.3**	54.0
celebrity & pop culture	22.8	22.9	22.3	**41.7**	34.4	**54.3**	47.2	**52.7**	50.5	**59.6**	43.7	**54.9**	**48.6**	47.8	46.4	**57.6**
diaries & daily life	18.2	**21.2**	25.8	**34.4**	**45.2**	44.0	46.2	**50.3**	43.5	**49.3**	44.6	**49.9**	44.5	**51.2**	44.7	**49.8**
family	3.5	**6.3**	33.9	**46.4**	47.2	**48.3**	50.6	**56.8**	52.8	**63.4**	46.1	**49.1**	46.4	**55.2**	53.1	**56.2**
fashion & style	**4.8**	4.1	38.4	**57.6**	52.8	**74.8**	66.4	**74.1**	66.4	**77.4**	56.0	**68.8**	**66.4**	75.2	67.2	**75.2**
film tv & video	**22.8**	22.0	47.3	**58.6**	62.8	**68.2**	64.4	**71.4**	64.7	**71.3**	66.8	**69.2**	66.1	**72.2**	65.4	**70.6**
fitness & health	6.6	**9.3**	35.7	**36.0**	**53.6**	52.2	52.4	**53.2**	62.4	**65.4**	**48.2**	38.7	**55.7**	42.2	58.6	52.6
food & dining	3.5	**4.6**	25.0	**41.7**	**70.1**	68.2	**75.1**	75.3	**79.3**	68.2	**74.5**	65.7	**75.4**	70.7	**80.4**	71.6
gaming	6.9	**7.5**	31.8	**45.0**	57.4	**61.2**	58.4	**61.4**	63.8	**69.1**	66.1	**67.6**	64.6	**69.2**	64.8	71.2
learning & educational	4.2	**4.5**	13.0	**13.9**	38.2	**43.2**	**49.5**	48.7	49.8	45.8	**42.9**	36.2	**49.3**	47.1	**48.9**	47.0
music	24.7	**25.5**	76.1	**81.8**	83.6	**86.0**	86.0	**87.1**	87.4	**88.1**	86.9	**87.2**	**88.1**	87.8	86.9	**88.2**
news & social concern	39.3	**39.9**	69.8	**76.9**	**83.8**	**83.8**	83.9	**84.6**	85.5	**85.9**	83.5	**84.3**	84.4	**86.2**	84.5	**85.0**
other hobbies	**10.5**	9.6	4.2	**15.0**	**27.0**	23.6	25.0	**28.4**	31.7	**35.4**	**23.1**	21.5	27.7	**30.3**	31.1	26.2
relationships	6.4	**7.3**	13.7	**36.3**	30.8	**35.2**	37.6	**51.8**	39.3	**56.8**	36.8	**51.2**	35.3	**51.6**	44.5	**54.0**
samples avg	13.8	**14.3**	57.0	**63.7**	**70.3**	72.0	73.1	**74.2**	74.4	**76.4**	**73.8**	73.2	74.3	**75.2**	74.7	**75.2**
science & technology	8.3	**9.3**	17.4	**35.8**	45.9	**50.3**	54.2	**56.4**	52.1	**59.4**	46.9	**53.2**	50.5	**56.0**	50.2	**52.1**
sports	**36.6**	34.8	82.2	**89.1**	93.1	**93.2**	**94.8**	94.2	94.6	**95.4**	**95.4**	94.4	**95.6**	94.8	**95.2**	94.8
travel & adventure	2.2	**3.5**	**17.7**	9.8	**21.7**	20.6	41.5	**47.7**	46.3	**59.9**	**38.5**	0.0	**57.1**	56.0	52.2	54.7
youth & student life	1.7	**2.9**	2.9	**12.4**	33.3	**44.6**	49.2	**52.4**	21.0	**46.0**	31.6	**35.2**	**50.4**	43.6	50.8	**51.0**
macro avg	12.6	**13.2**	30.5	**41.5**	50.1	**54.6**	55.8	**60.3**	56.3	**63.0**	52.7	**53.1**	57.2	**59.6**	58.8	**60.9**

Table 3.5: Macro average F1 scores for the multi-label setting when using temporal (temp) and random (rand) split. Highlighted with bold is the best score for each model.

We continue our analysis by investigating in more detail TimeLM-19's results, which is the best performing model according to the evaluation (Section 3.4). Figure 3.4 displays the TimeLM-19 performance differences between the temporal and random splits on the single-label setting. In general, results are overall better in the random split, with an overall relative decrease of 4.3% in Macro-F1 for the temporal split.

The largest decrease in performance is observed for the *arts & culture* topic in both settings, which can be attributed to a fast evolving vocabulary. In contrast, *business & entrepreneurs* does not see any decreased in performance in both settings, and results are even slightly better on the temporal split.[14]

[14]In the Appendix **??** we provide a detailed analysis by quarter, in order to better understand the temporal aspect. The results confirm how the performance of *arts & culture* decreases over time, while for the rest of the topics the trend is unclear.

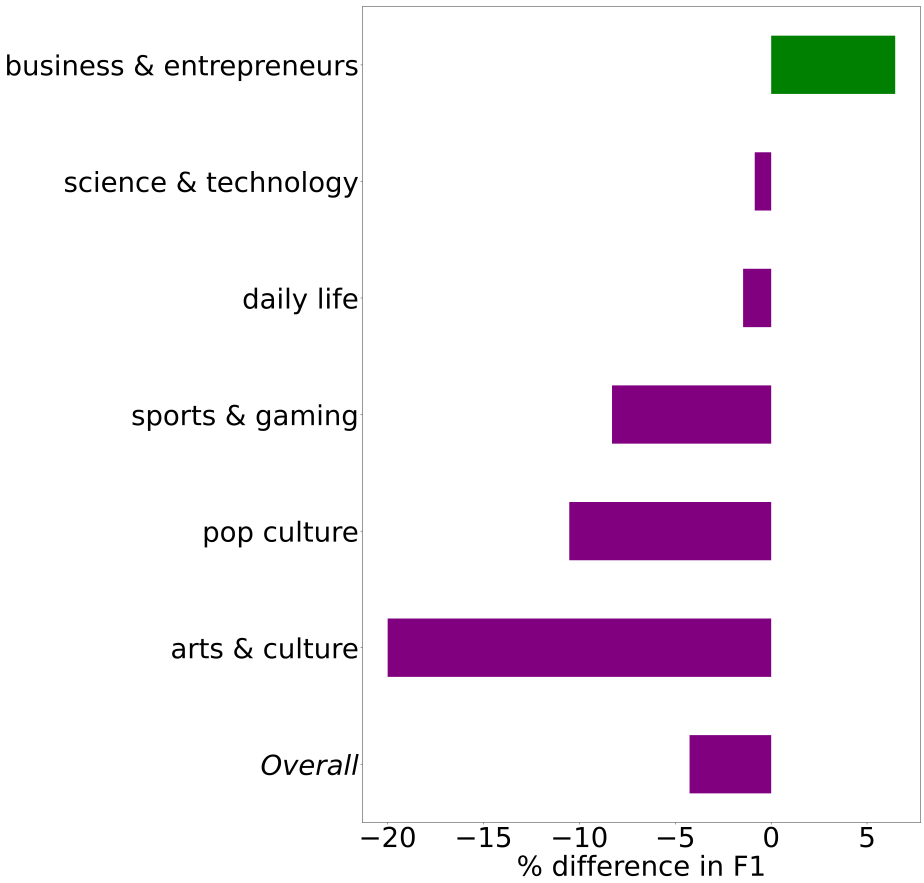

Figure 3.4: Relative (%) differences in F1 scores when *TimeLM-19* is trained in a temporal and in a random setting for the single-label setting. Negative values indicate that when using the temporal split the model's performance decreases.

3.5.2 Error analysis

To better understand the nature of errors made by language models, Figure 3.5 shows a confusion matrix for the best-performing TimeLM-19 model in the single-label setting. The model seems to struggle with tweets assigned to the *arts & culture* topic with 68% of them being misclassified as *daily life*. These errors include entries such as "Happy Day of the Dead 2020! #GoogleDoodle" or "Gifts of love are the ingredients of a #MerryChristmas Give your loved ones a physical/virtual

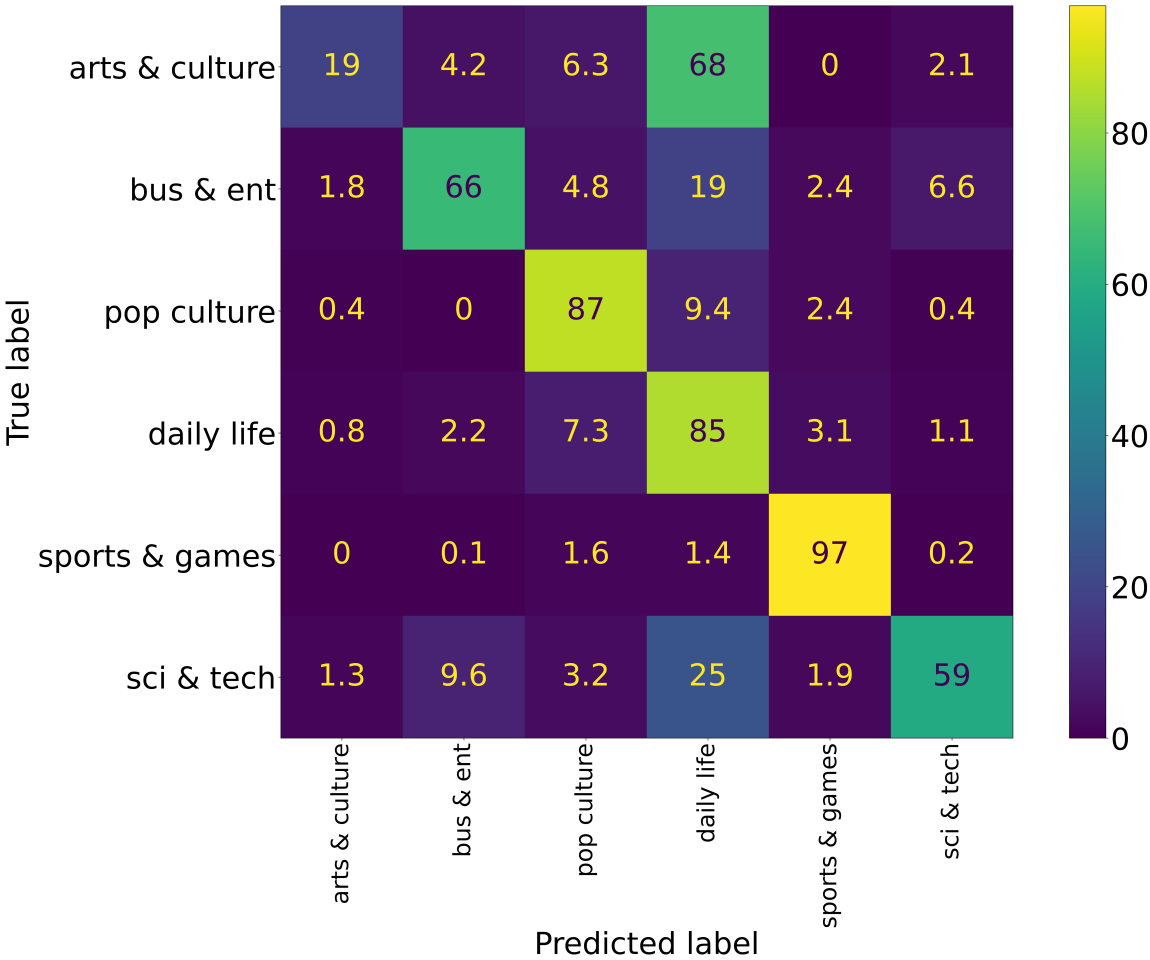

Figure 3.5: Confusion matrix of the TimeLM-19 results for the single-label setting. The values displayed are normalized by row.

crypto gift card within the {{USERNAME}} app". While these tweets revolve around religious/cultural holidays, one might also associate them to daily life events, which also shows the challenging nature of this dataset.

Another topic that is frequent misclassified is *science & technology*, with 41% of the tweets being assigned to the wrong topic. When looking at the errors we identify tweets such as "Bill Gates-Funded Company Releases Genetically Modified

Mosquitoes in US", classified as *business & entrepreneurs*, and "Monday's Google Doodle Celebrates Jupiter And Saturn On The Winter Solstice via Forbes", classified as *daily life*.

In other cases, further investigation would be required to understand the source of the mistakes, e.g., "A year ago we looked at PE10s across the world on URL The latest Weekly Macro Themes takes a look at how the Euro Area stacks up now." was classified as *sports* instead *business & entrepreneurs*. The nature of these types of error, as well as the relatively low performance of models compares to other topic classification datasets, suggest that there is ample room for improvement.

When considering the multi-label setting, there are topics with high percentage of errors such as *celebrity & pop culture* and *diaries & daily life*. There are entries like "Anyone else notice {O Shea Jack Nichol son} hasn't tweeted about the Lakers making the conference finals? Weird. You good man?" where the model correctly classifies it as *sports* but fails to classify it as *celebrity & pop culture*, being probably unaware of the celebrity status of the person being mentioned. The *diaries & daily life* topic seems to be particular confusing for the model and fails to identify it in tweets such as "Lost all my bets on the Kentucky Derby today but scored a tee time at {{USERNAME}} Black course next weekend I'd say I came out a winner.", and "Faceing difficulty while login to internet banking for the 1st time using Id and password provided in the welcome kit didn t expected this from such a good bank {Canara Bank}", even though they are correctly assigned the *sports* and *business & entrepreneurs* topics, respectively.

3.6 Summary

In this chapter we presented TweetTopic, the first large-scale dataset for tweet topic classification. Given the prominence of social media in recent times, this dataset can help build supervised models for clustering and organising the online content.

The curated set of topics contains a diverse and broad set of categories that cover most topics present in social platform data. This dataset can further motivate research on the evolution of these initial topics on social platforms, i.e., the extension of the existing categorization to new topics or subtopics that will emerge and fade over time due to user engagement.

Finally, TweetTopic has been shown to be relatively resilient to temporal changes, and it offers easily interpretable results. Based on these contributions, we believe that this dataset will be useful for a significant number of researchers and practitioners working on social media, including Computational Social Science and Data Mining experts, given the relevance of the topic for extracting information and understanding online behaviour.

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